



AI4VIS: AI-POWERED INTERACTIVE CREATION OF VISUALISATIONS - STANDARD CHARTS

Final Year Project Presentation

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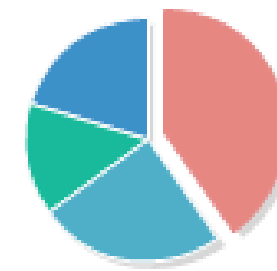
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Standard Charts

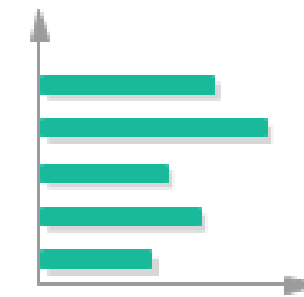
- Standard charts are commonly used visualizations for representing structured data.
- They are designed for clear and easy interpretation.

Types of Standard Charts

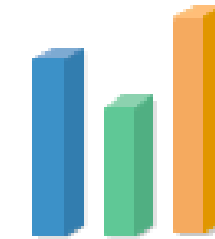
- Bar Chart
- Line Chart
- Pie Chart
- Scatter Plot



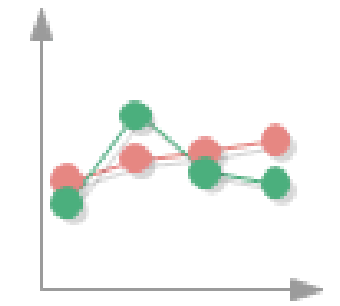
Pie



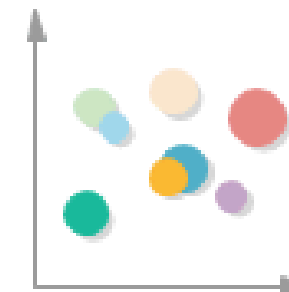
Bar



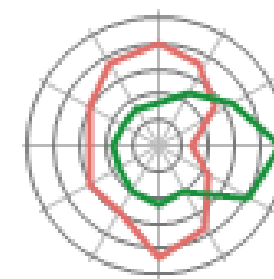
Column



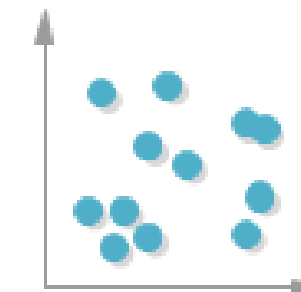
Line



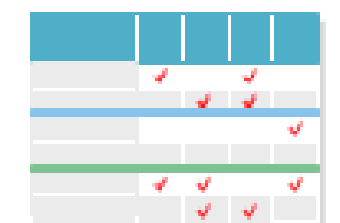
Bubble Chart



Spider and Radar



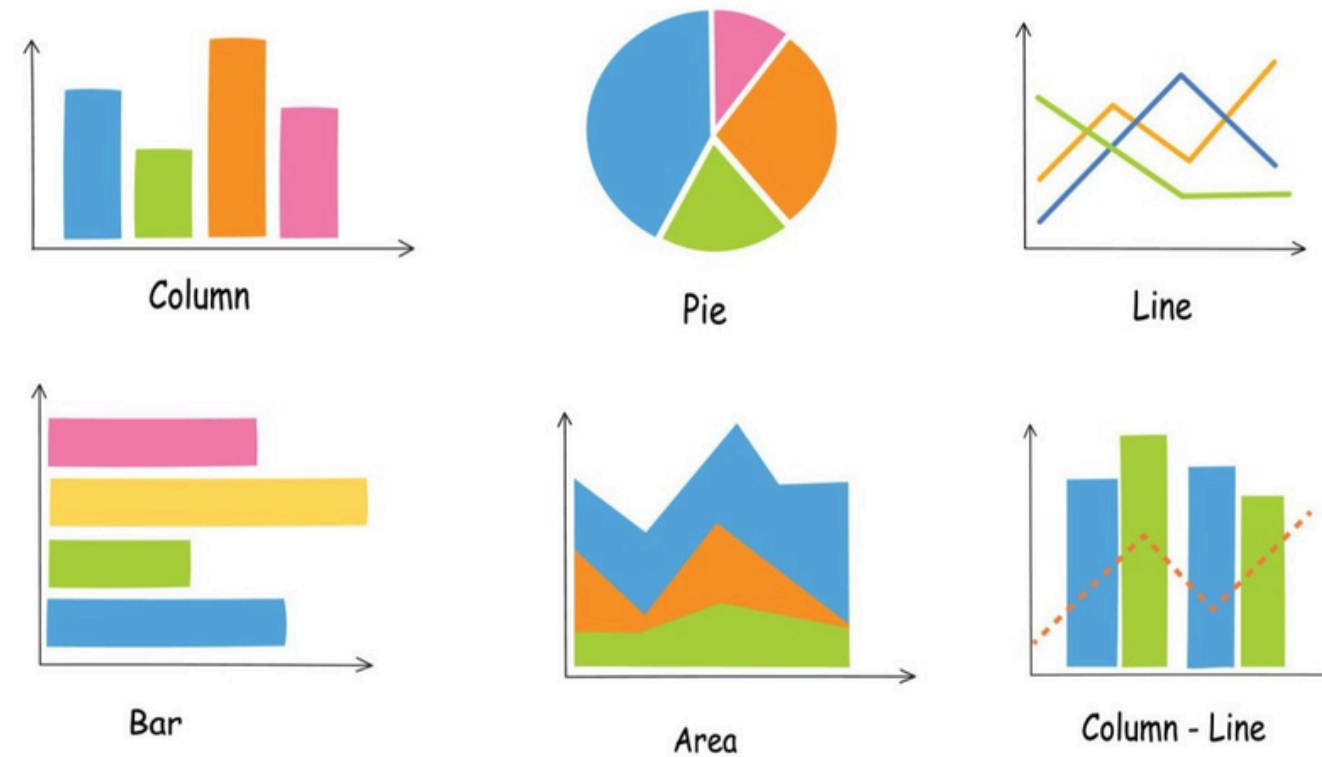
Scatter



Comparison Chart

Introduction

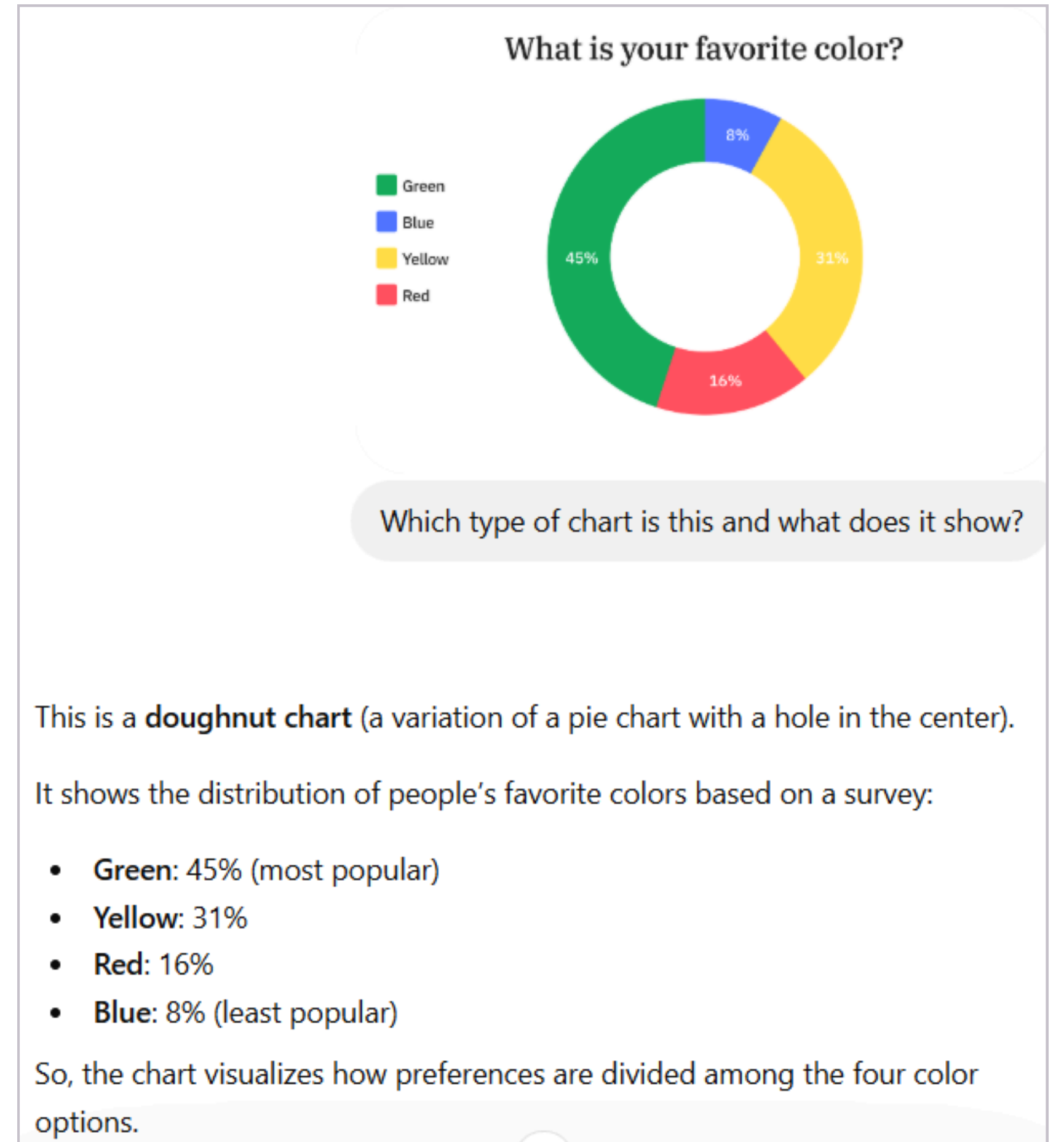
Data visualization is the process of representing data in graphical form to communicate information clearly and effectively.



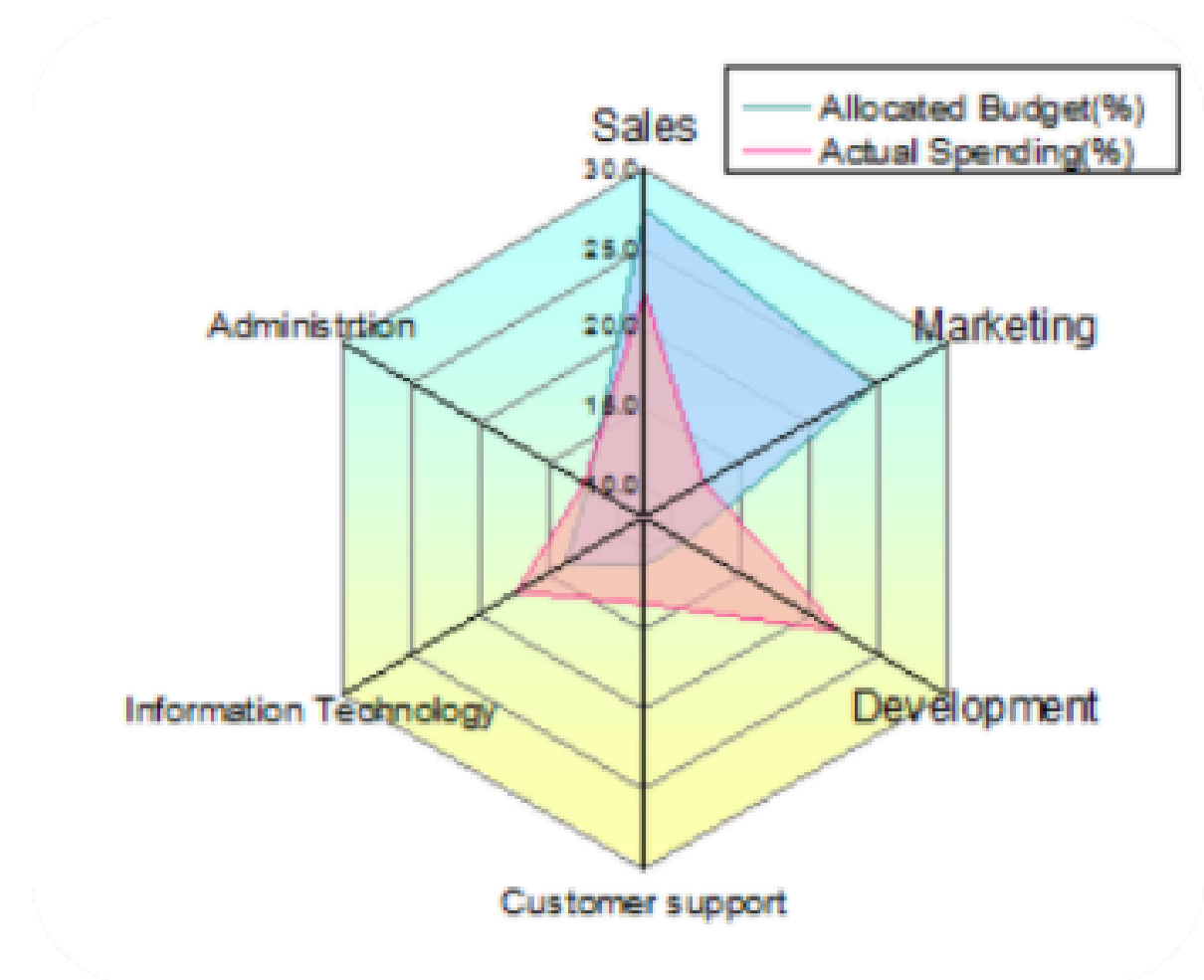
Charts are everywhere!

Introduction

- **Large Language Models (LLMs)** are increasingly being used for data analysis and visualization tasks.
- LLMs are capable of supporting data-related tasks, including generating, interpreting, and reasoning over visual representations through natural language interaction



Example of Donut chart provided to ChatGPT

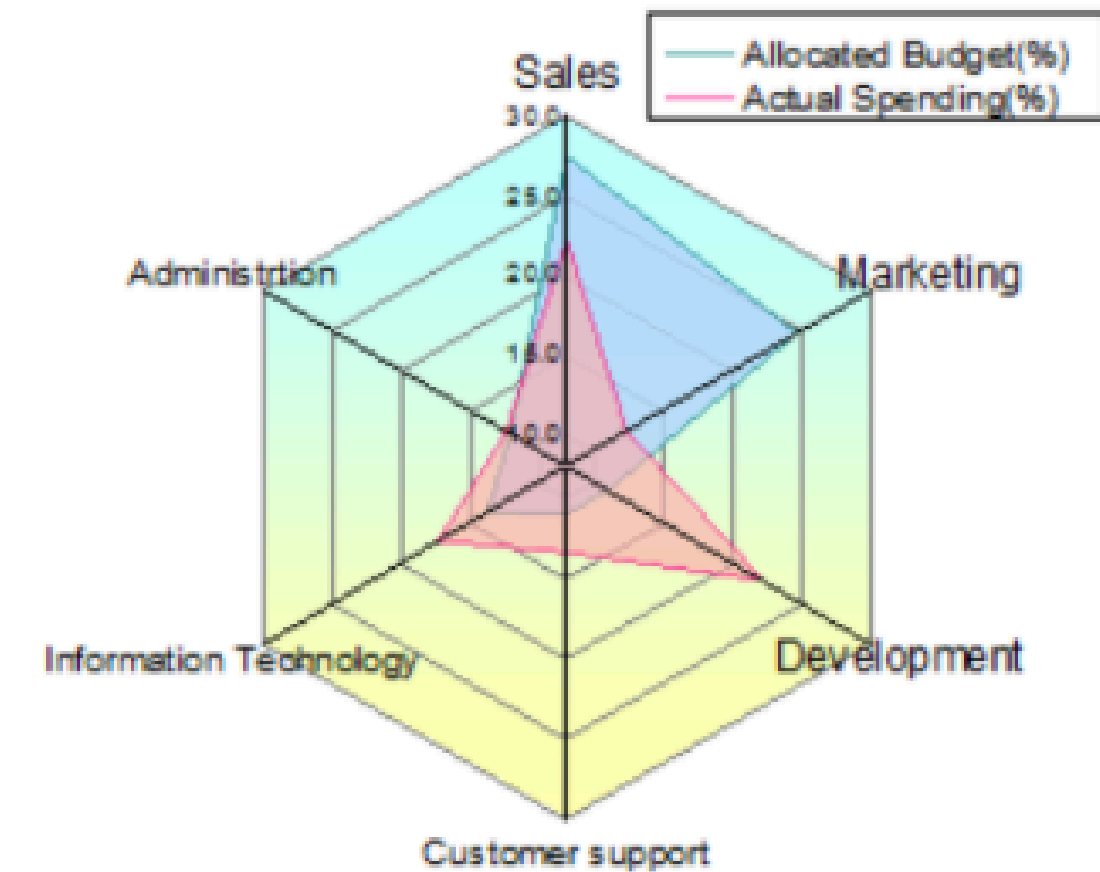


Which category has the lowest actual spending percentage?

From the chart's pink "Actual Spending (%)" values, **Information Technology** is the lowest (around 6%), lower than Customer Support (~8%) and all other categories.

Example of Radar chart provided to ChatGPT

Given Output: Information Technology
Expected Output: Administration



Which category has the lowest actual spending percentage?

From the chart's pink "Actual Spending (%)" values, **Information Technology** is the lowest (around 6%), lower than Customer Support (~8%) and all other categories.

Example of Radar chart provided to ChatGPT

Problem Statement

- The reliability of LLMs in handling chart-related tasks remains an open research challenge.
- Limitations in these areas may lead to incorrect interpretations, flawed insights, and the potential propagation of misleading information.

How well can LLMs actually understand and reason about charts?

Objectives

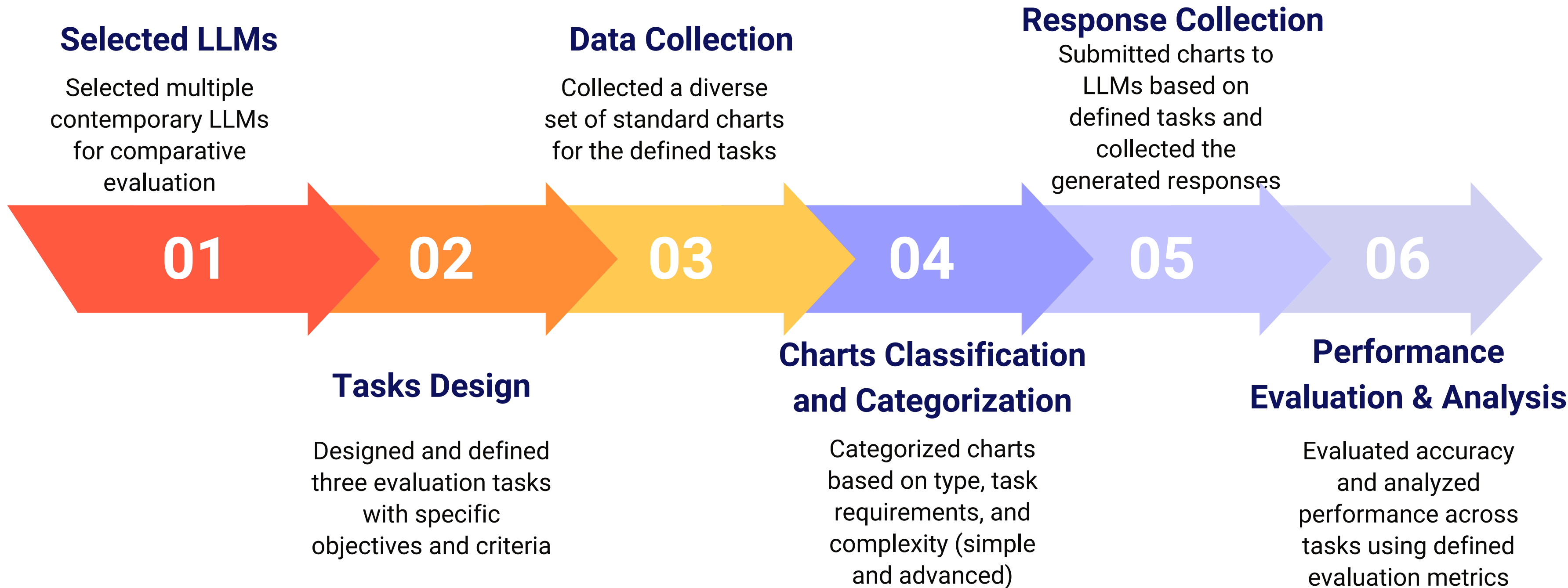
- To evaluate and compare the performance of selected commercial and open source LLMs across multiple chart-related task types.
- To analyze common error patterns exhibited by LLMs when interpreting and reasoning about charts.
- To identify the strengths and limitations of each evaluated LLM in handling graphical reasoning tasks

Literature Review

- Recent work demonstrates their potential for visualization applications: LLM4Vis showed that LLMs can produce visualization specifications from prompts
- Despite these advances, LLMs still struggle with accuracy and consistency, especially for complex charts or multi-dimensional visualizations
- Studies like ChartQA have highlighted that models often make errors in extracting values, comparing data points, and reasoning about visual encodings

This motivates a systematic evaluation of multiple LLMs across diverse chart-related tasks, aiming to assess the accuracy, reliability, and performance of these models in data reasoning and analysis

Methodology



Selected Large Language Models (LLMs)

LLMs	Version/Model	Developed By
ChatGPT	Model GPT-5.2	OpenAI
Claude	Model Claude Sonnet 4.5	Anthropic
Gemini	Model Gemini 2.5 Flash	Google
Mistral	Model Mistral-Large-2407	Mistral AI

These models are chosen due to their popularity, accessibility, and claimed capabilities in data reasoning and analysis.

Charts Categorization

To systematically evaluate LLMs performance across different tasks, the charts used in this study are categorized based on their structural complexity and common usage in practice.

The charts in this study are organized into two categories:

1. **Common Charts:** which are widely used and relatively straightforward to interpret
2. **Advanced Charts:** which feature more complex layouts or multi-dimensional encodings

This classification allows for a comparative evaluation of LLMs performance on both simple and challenging visualizations. This helps identify strengths and limitations of LLMs and supports systematic and structured evaluation

Tasks Overview

Evaluation is divided into multiple tasks targeting different aspects of data visualisation and analysis. These tasks assess chart interpretation, reasoning, and understanding of visual information across different complexity levels.

They include applying visual encoding principles, extracting data from charts, and identifying misleading visualisations, enabling a comprehensive evaluation of model capabilities.

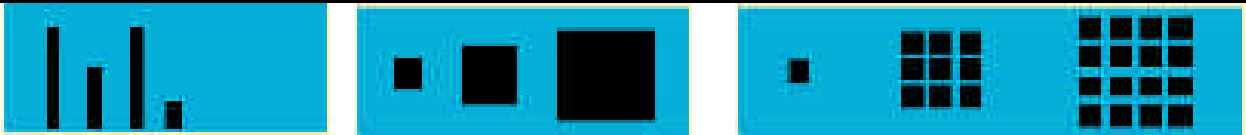
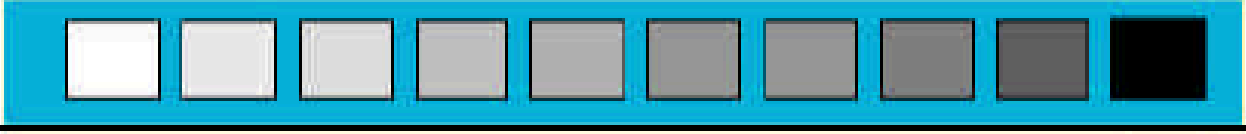
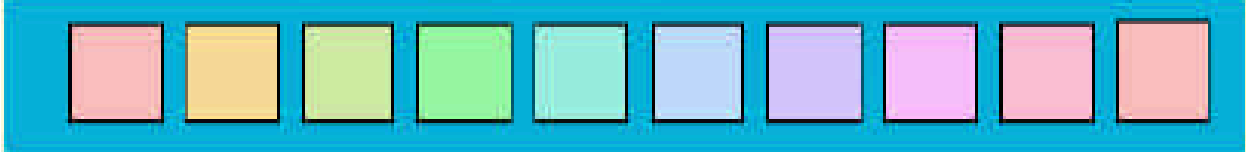
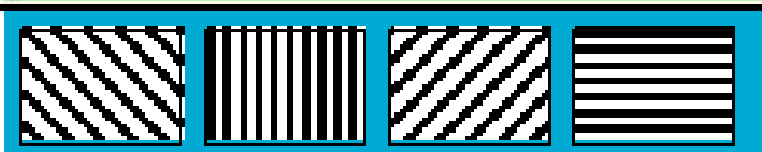
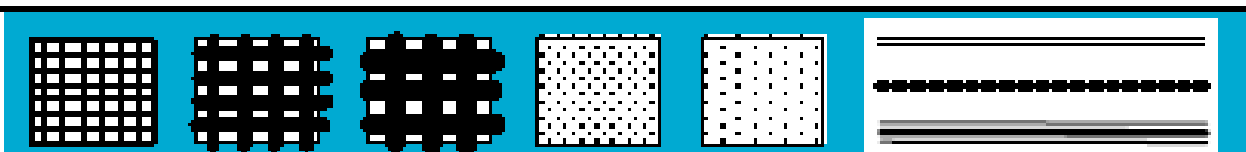
Task 1 Visual Encoding Interpretation	Task 2 Chart-Based Question Answering	Task 3 Detection of Misleading Charts
• Interprets visual encoding elements such as color, position, shape, and length. • Focuses on basic chart understanding and interpretation accuracy.	• LLMs are required to answer questions based on chart data, including values, trends, and comparisons. • Focuses on accuracy and consistency in data interpretation and reasoning.	• Evaluates the ability of LLMs to identify and analyze flawed or misleading charts by detecting issues such as incorrect scales and inappropriate use of colors. • Focuses on LLMs ability to detect misleading visualisation.

Task 1: Visual Encoding Interpretation

Visual Encoding is the way data is represented visually in a chart or graph using visual properties instead of raw numbers. It is how information is translated into visual form so that humans (or models) can understand patterns and relationships easily.

Visual Encoding Elements:

These are the visual properties used to represent data:

Bertin's Original Visual Variables	
Position changes in the x, y location	
Size change in length, area or repetition	
Shape infinite number of shapes	
Value changes from light to dark	
Colour changes in hue at a given value	
Orientation changes in alignment	
Texture variation in 'grain'	

Task 1: Visual Encoding Interpretation

This task evaluates the ability of LLMs to interpret visual encoding elements in standard charts.

Objective: To assess how accurately LLMs can identify, understand and reason about visual encoding elements when interpreting charts. It evaluates how well LLMs understand different ways data is visually represented.

Chart Categorization

The follows charts are used in this task:

S.no	Common Chart	Advanced Chart
1	Bar Chart	Radar Chart
2	Line Chart	Stream Graph
3	Histogram	Tidy Tree
4	Scatter Plot	Treemap
5	Pie Chart	Candlestick
6	Donut Chart	Sankey Diagram
7	Box Plot	Parallel Coordinates
8	Heatmap	Sunburst Chart
9	Stacked Area Chart	Geospatial Chart
10	Grouped Bar Chart	Network Graph

Visual Encoding Elements Considered

This task focuses on four key visual encoding elements:

S.no	Category
1	Color Hue
2	Position
3	Shape
4	Length

1. Color (Hue)

- Identifies colors used for categories
- Maps color → correct data group

2. Position

- Interprets data placement on axes
- Understands x-axis vs y-axis

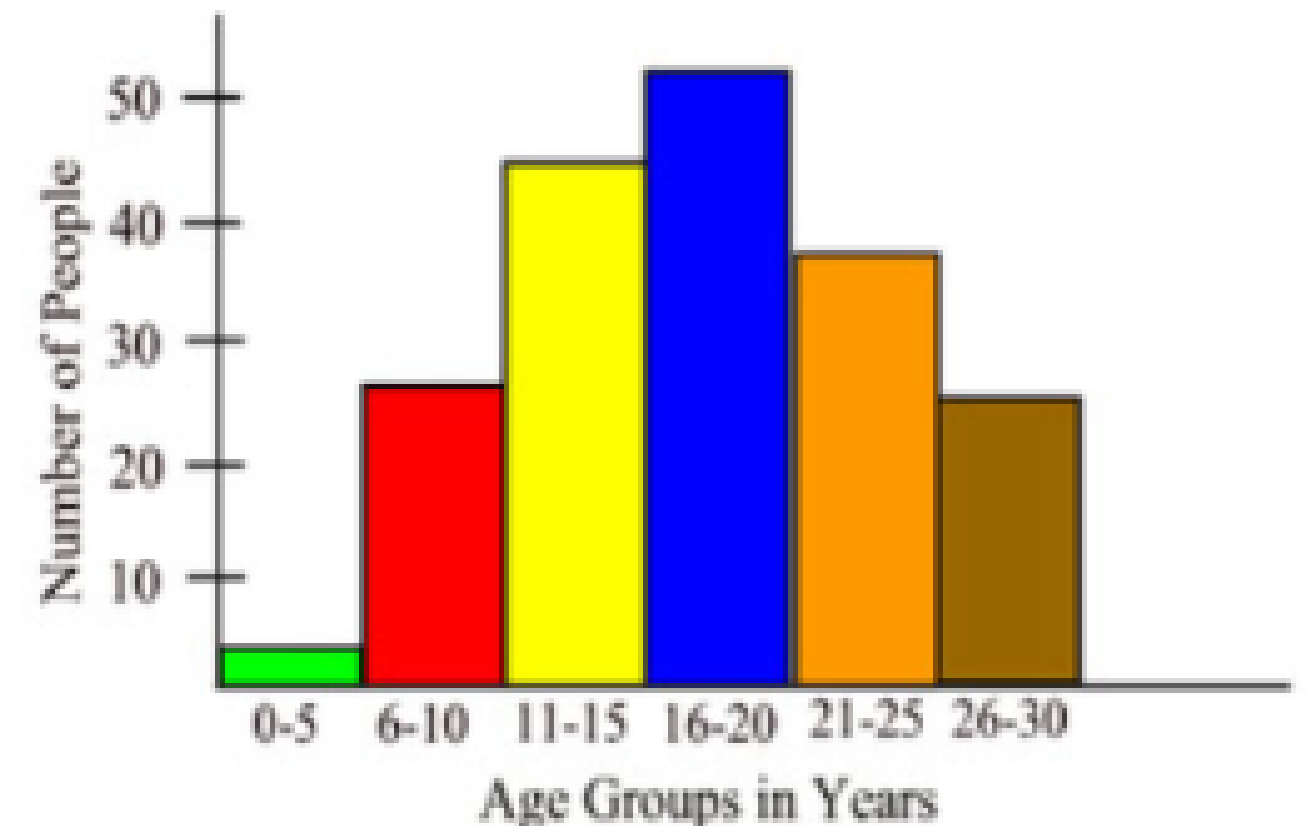
3. Shape

- Recognizes visual forms (bars, points, markers)
- Links shapes to data groups

4. Length

- Compares values using size/height
- Interprets bar/chart magnitude

Frequency Histogram



Which color represents the 16–20 age group?

Which axis shows the number of people?

What shape is used to represent each age group?

What is the length of the bar representing the age group with the second-highest value?

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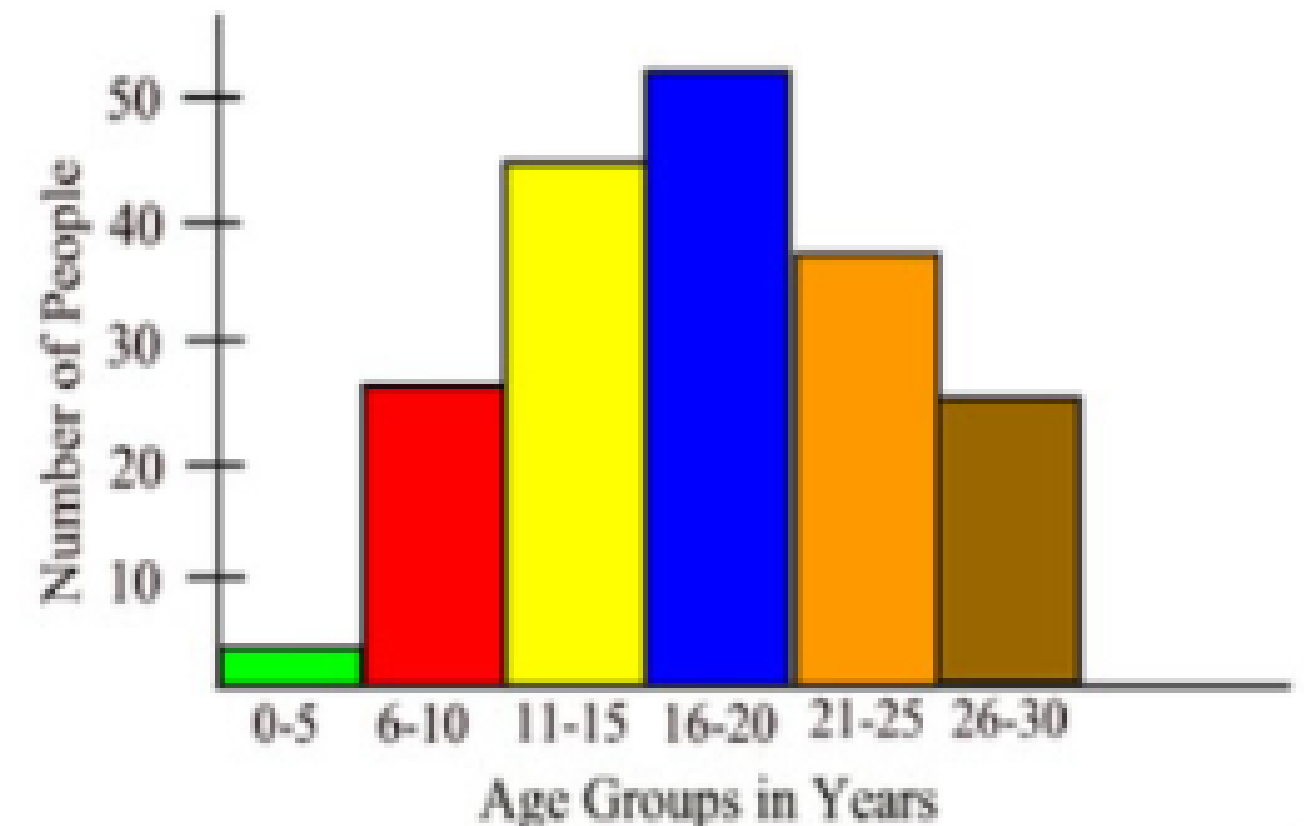
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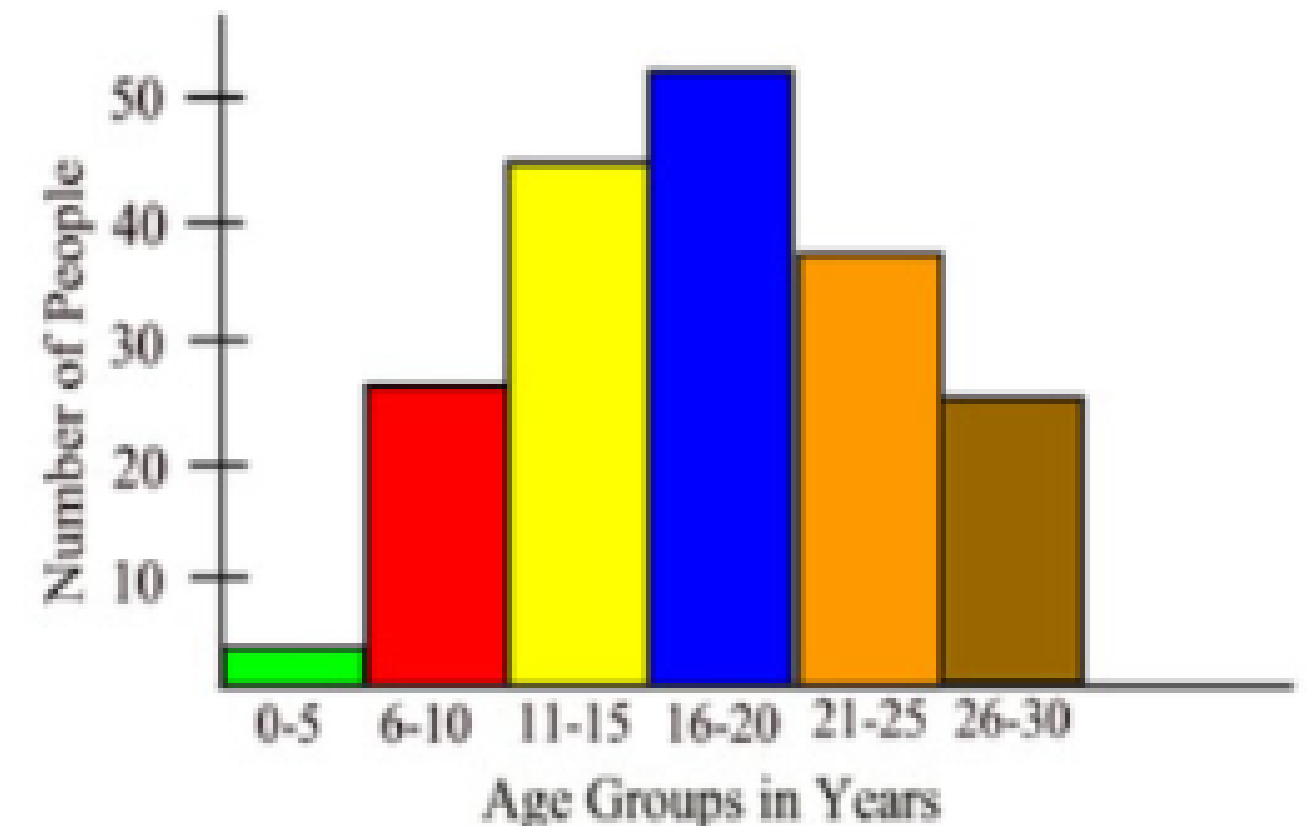
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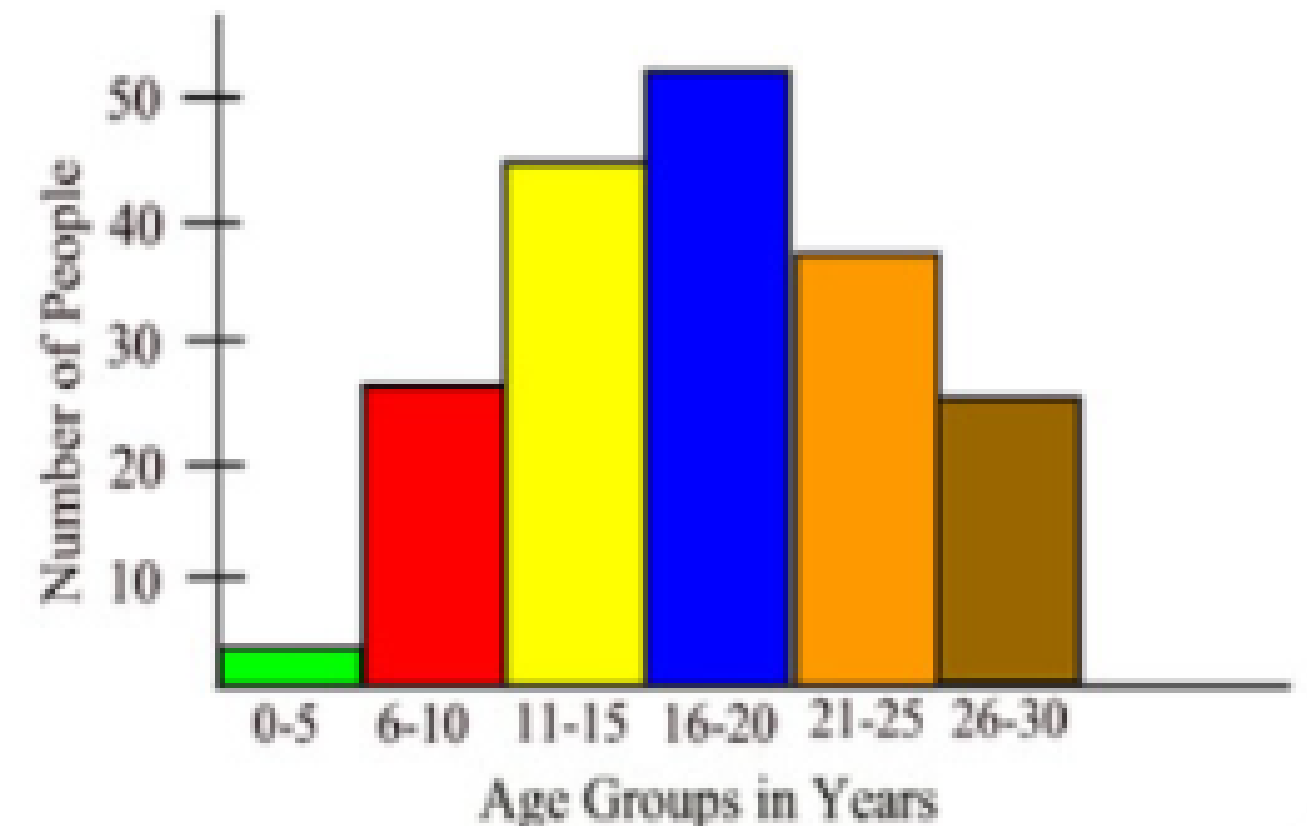
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Task 2: Chart-Based Question Answering

This task evaluates the ability of LLMs to perform analytical reasoning using chart images.

Objective: To whether models can extract insights, compare values, and reason about data represented in charts. LLMs are provided with chart images and asked structured questions to evaluate their understanding of values, trends, and relationships within the data.

Chart Categorization

The follows charts are used in this task:

S.no	Common Chart	Advanced Chart
1	Bar Chart	Radar Chart
2	Line Chart	Stream Graph
3	Histogram	Tidy Tree
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7	Box Plot	Parallel Coordinates
8	Heatmap	Sunburst Chart
9	Stacked Area Chart	Geospatial Chart
10	Grouped Bar Chart	Network Graph

Question Categories

Each chart is evaluated using structured question types:

S.no	Category
1	Highest Value Identification
2	Lowest Value Identification
3	Direct Value Retrieval
4	Similarity Detection
5	Ranking
6	Difference Between Values
7	Counting-Based Reasoning

1. Highest Value Identification

- Identifies maximum value represented in a chart

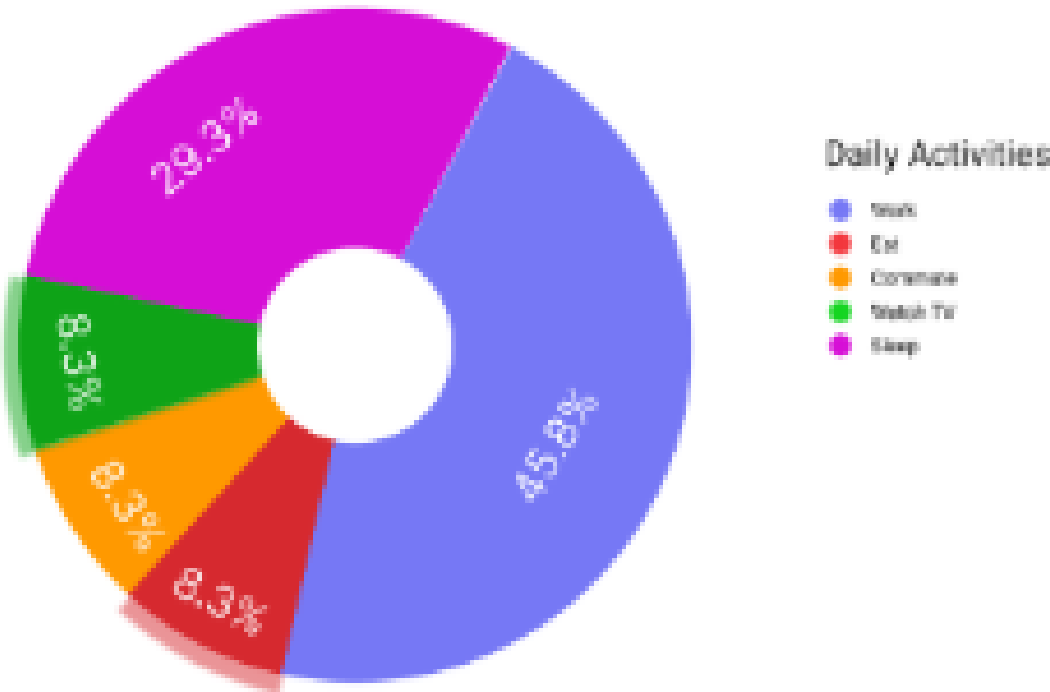
2. Lowest Value Identification

- Identifies minimum value represented in a chart

3. Direct Value Retrieval

- Extracts specific values directly from a chart

Doughnut Chart | Donut chart UI



Which daily activity has the highest percentage in the chart?
Which activity has the lowest percentage?
What percentage of time is spent watching TV?
Are any activities equal in percentage?
What is the rank of "Sleep" among all activities?
What is the difference between the highest and lowest activity percentages?
How many activity have more than 20%?

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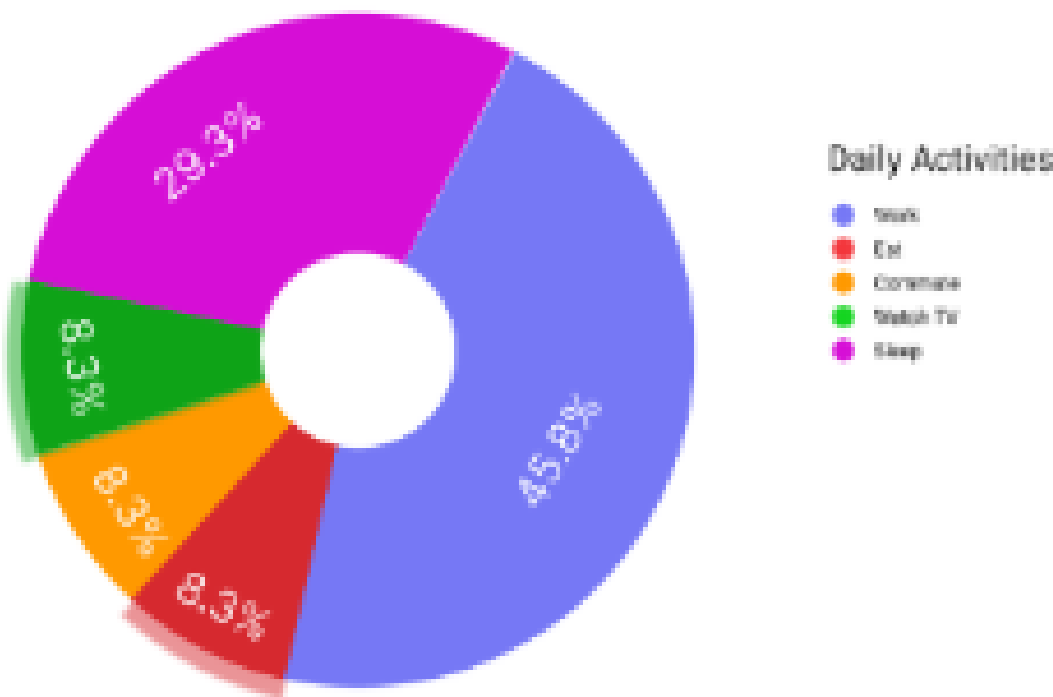
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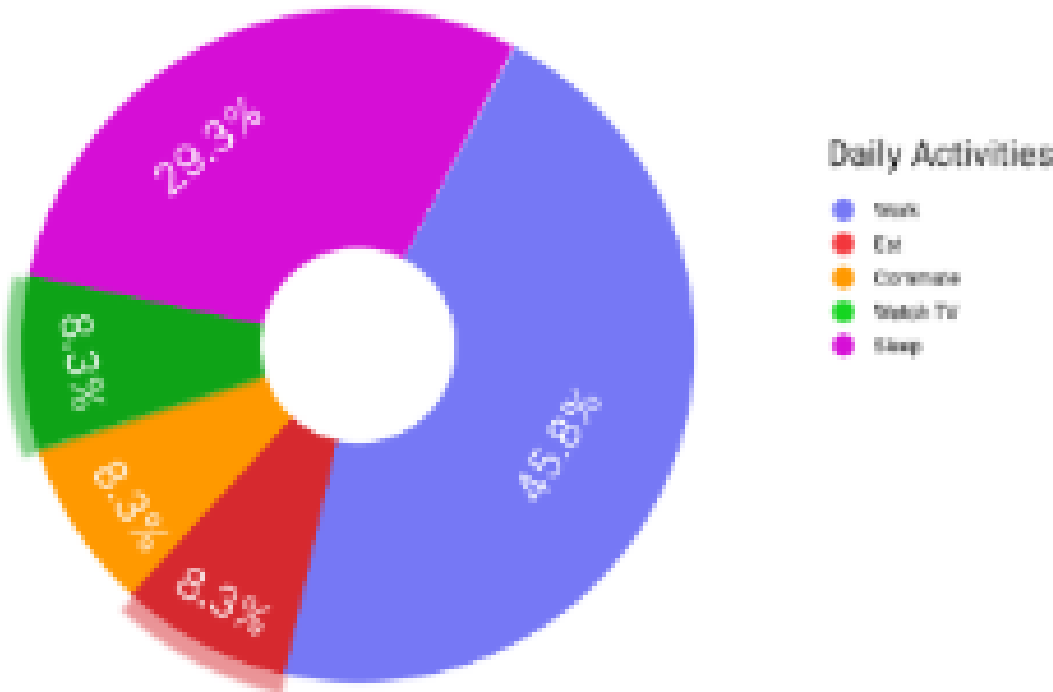
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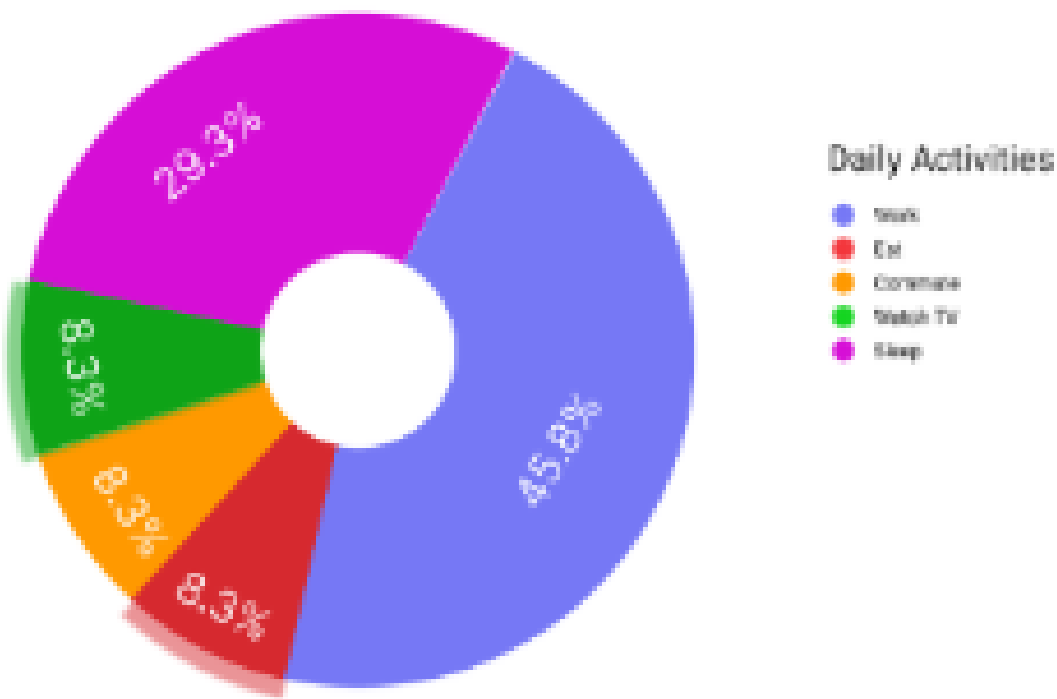
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Doughnut Chart | Donut chart UI



4. Similarity Detection

- Identifies data points with similar or closely related values.

5. Ranking

- Orders data points based on their values.

6. Difference Between Values

- Computes or estimates the difference between two data points

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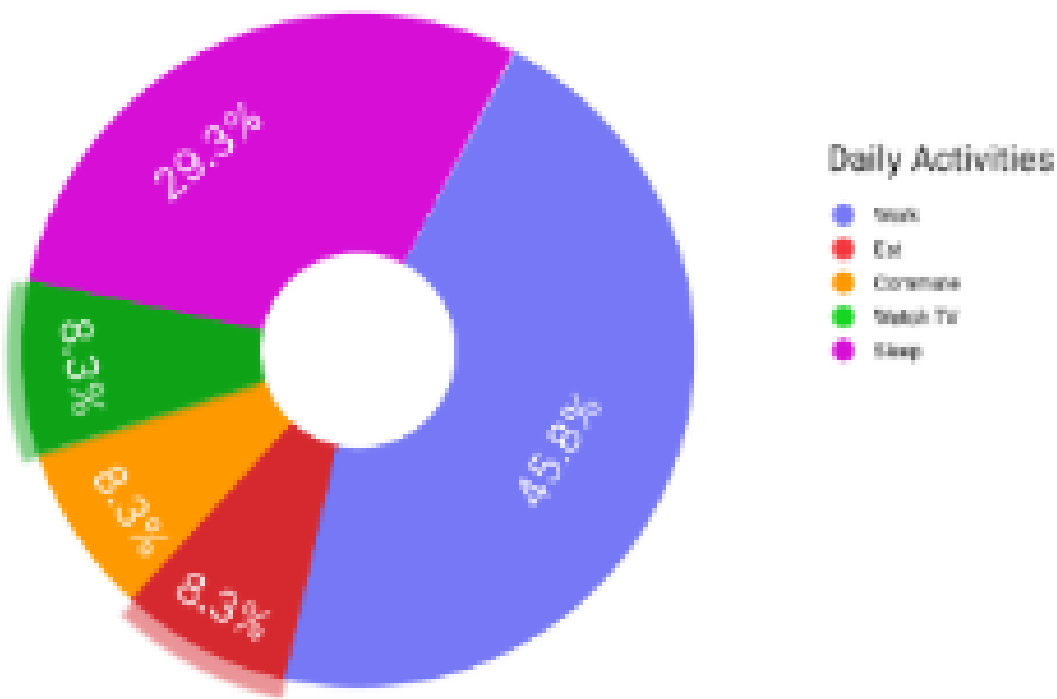
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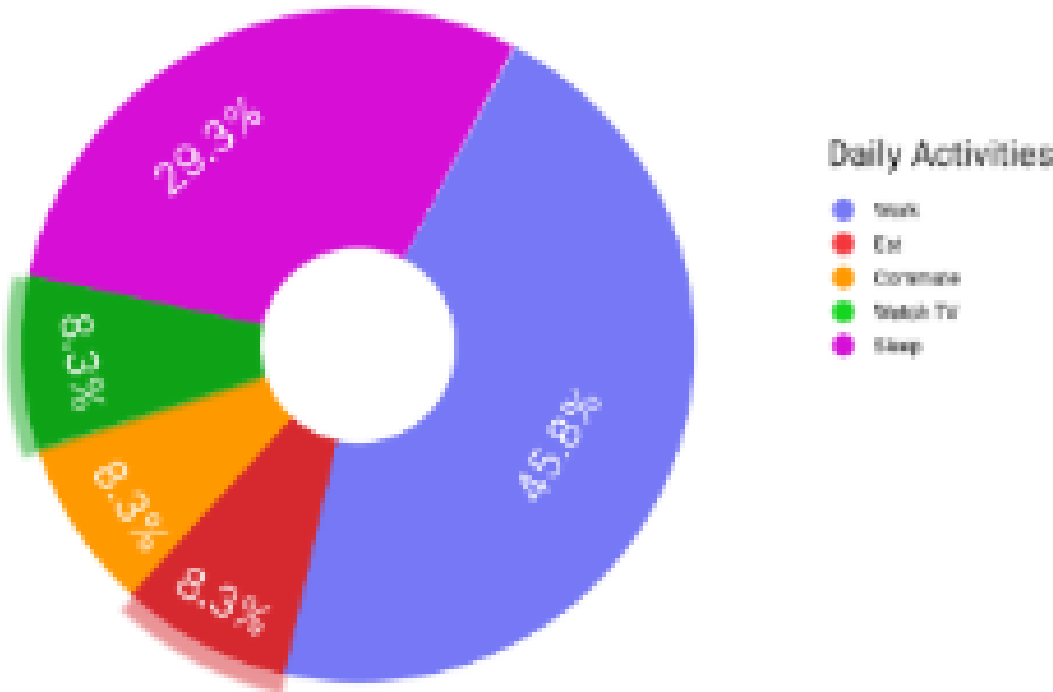
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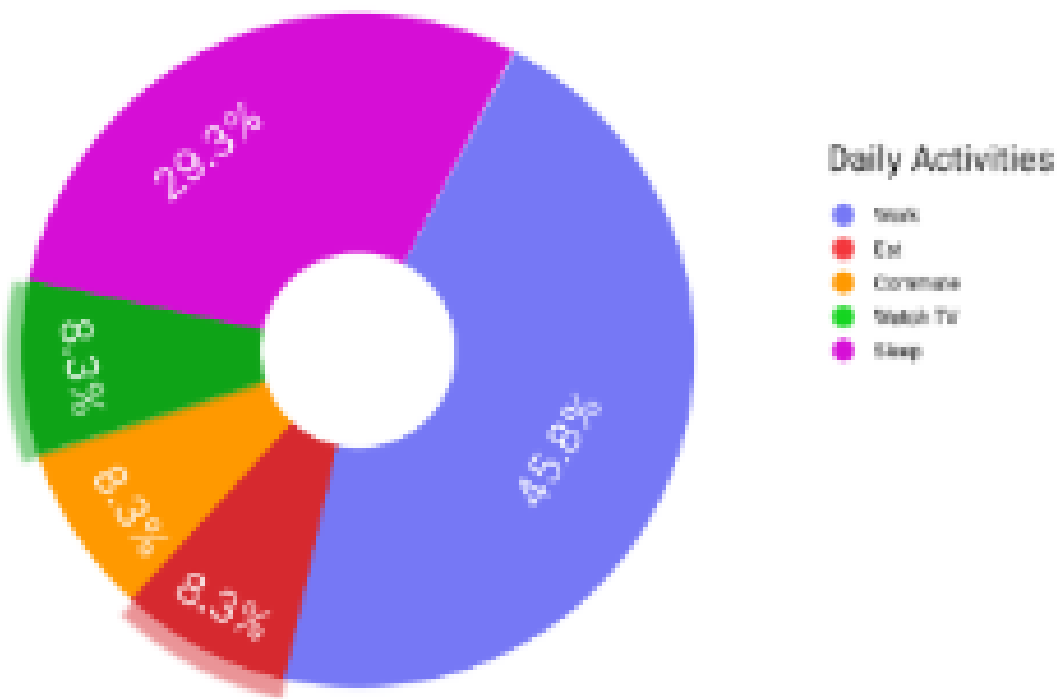
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Doughnut Chart | Donut chart UI



7. Counting-Based Reasoning

- Counts the number of data points that satisfy a given condition.

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What percentage of time is spent watching TV?
Are any activities equal in percentage?
What is the rank of "Sleep" among all activities?
What is the difference between the highest and lowest activity percentages?
How many activity have more than 20%?

Task 3: Detection of Misleading Charts

This task evaluates the ability of LLMs to identify and explain misleading or incorrectly designed visualizations.
The charts used in this task are intentionally designed with common visualization flaws to test the models' critical reasoning ability.

Objective: This task focuses on evaluating whether LLMs can assess the correctness and reliability of charts based on visualization principles..

Chart Categorization

The follows charts are used in this task:

S.no	Common Chart	Advanced Chart
1	Bar Chart	Radar Chart
2	Line Chart	Stream Graph
3	Histogram	Treemap
4	Scatter Plot	Sankey Diagram
5	Pie Chart	Parallel Coordinates
6	Donut Chart	Sunburst Chart
7	Heatmap	Geospatial Chart
8	Stacked Area Chart	Network Graph
9	Grouped Bar Chart	
10	Stacked Bar Chart	

Misleading Charts Categories

Each category evaluates different aspects of flaws in misleading visualisations:

S.no	Category
1	Choice of Chart
2	Color Mess
3	Overplotting / Overcrowding
4	Incorrect Size Encoding
5	3D Distortion
6	Missing / Ambiguous Legend



1. Choice of Chart

- Identify the inappropriate use of chart types for representing given data.

Are there any flaws in the chart? Detect what is wrong and explain why.

Misleading Charts Categories

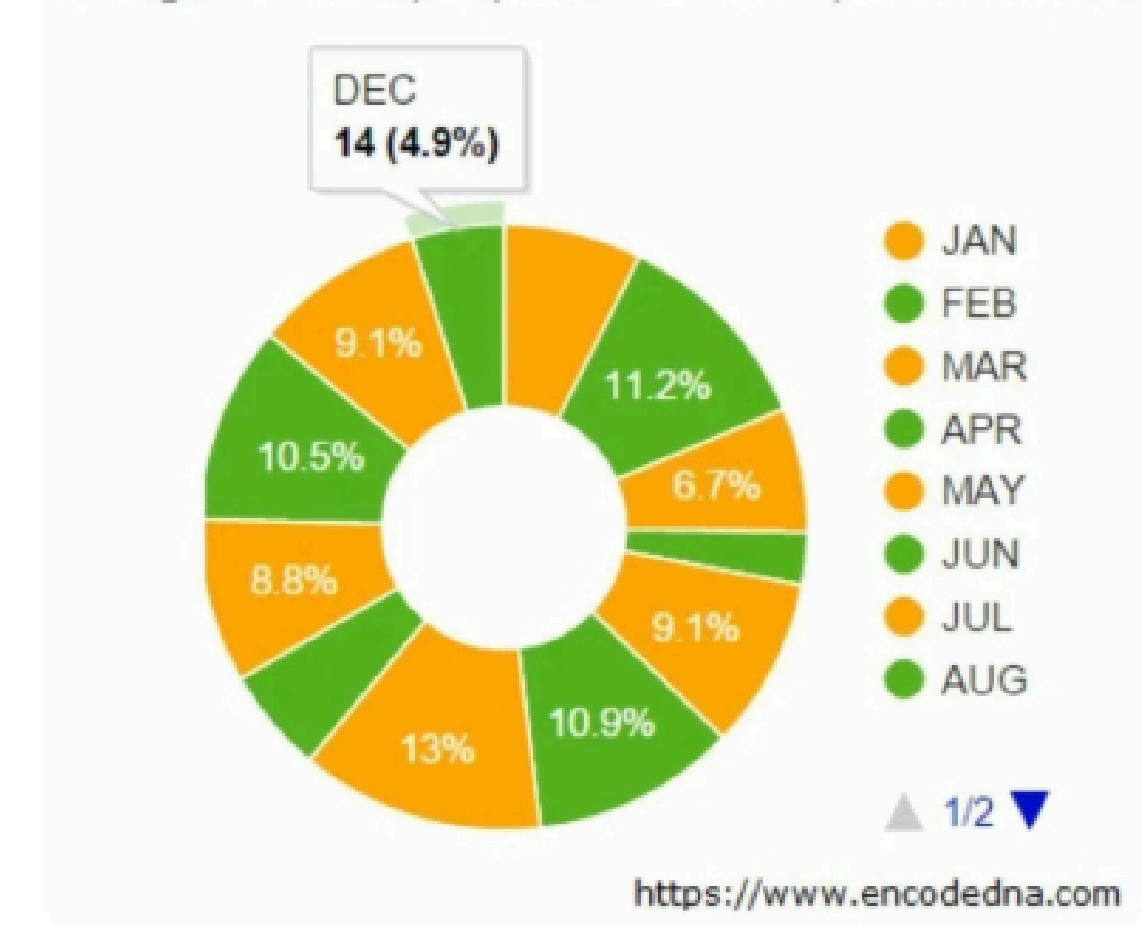
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2. Color Mess

- Detects poor or misleading use of color, such as excessive colors, inconsistent color mapping etc.

Google Donut (Pie) Chart with Dynamic Data

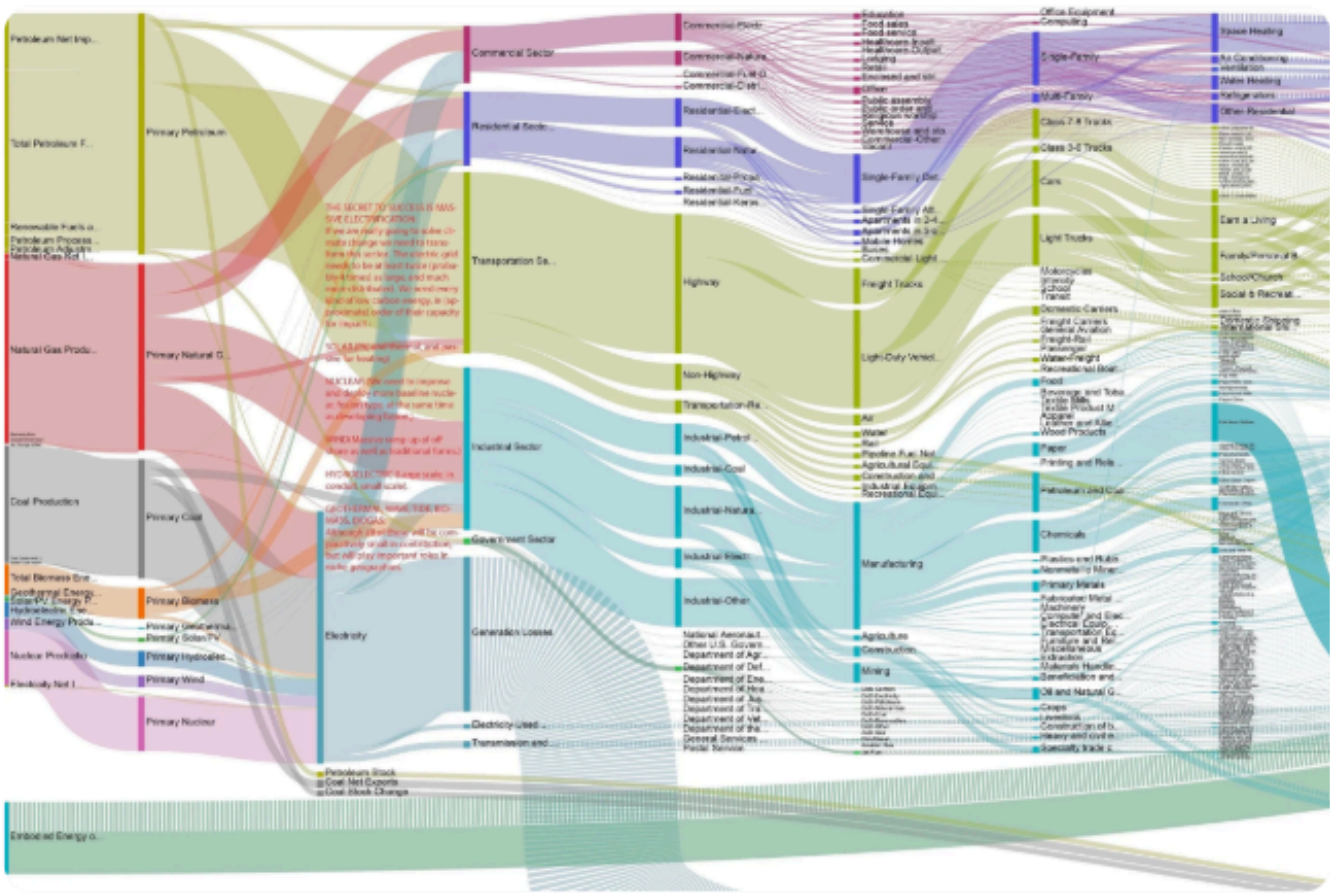


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3. Overplotting / Overcrowding
- Recognizes when a visualization becomes difficult to interpret due to excessive information.

Are there any flaws in the chart? Detect what is wrong and explain why.

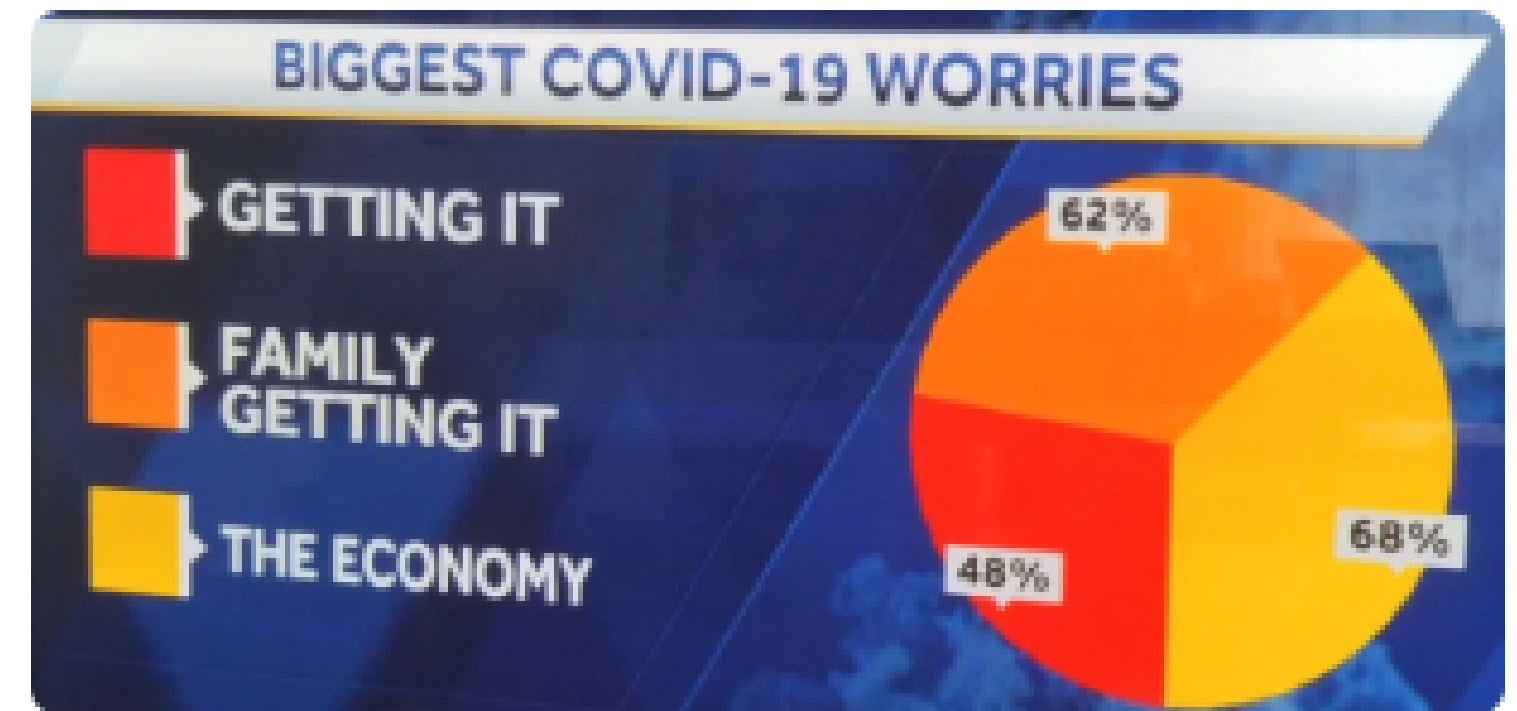
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4. Incorrect Size Encoding

- Identify mismatches between the data and its visual encoding.



Are there any flaws in the chart? Detect what is wrong and explain why .

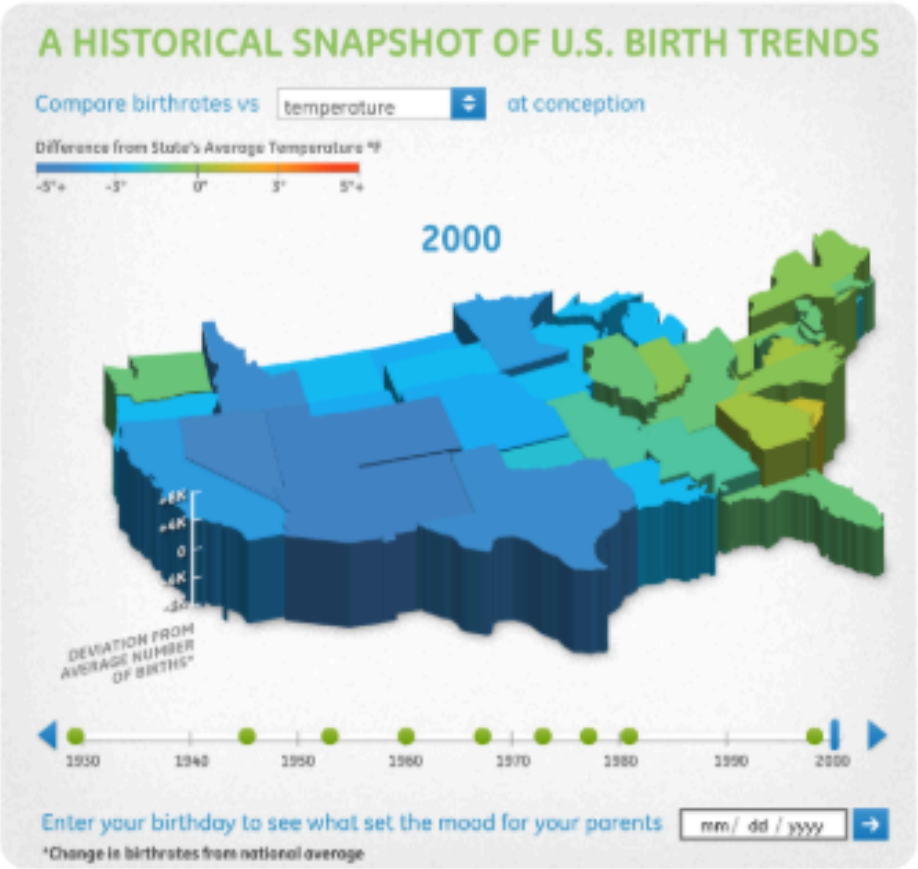
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6	Missing / Ambiguous Legend

5. 3D Distortion

- Recognizes how 3D visualizations can introduce bias or misinterpretation.



Are there any flaws in the chart? Detect what is wrong and explain why.

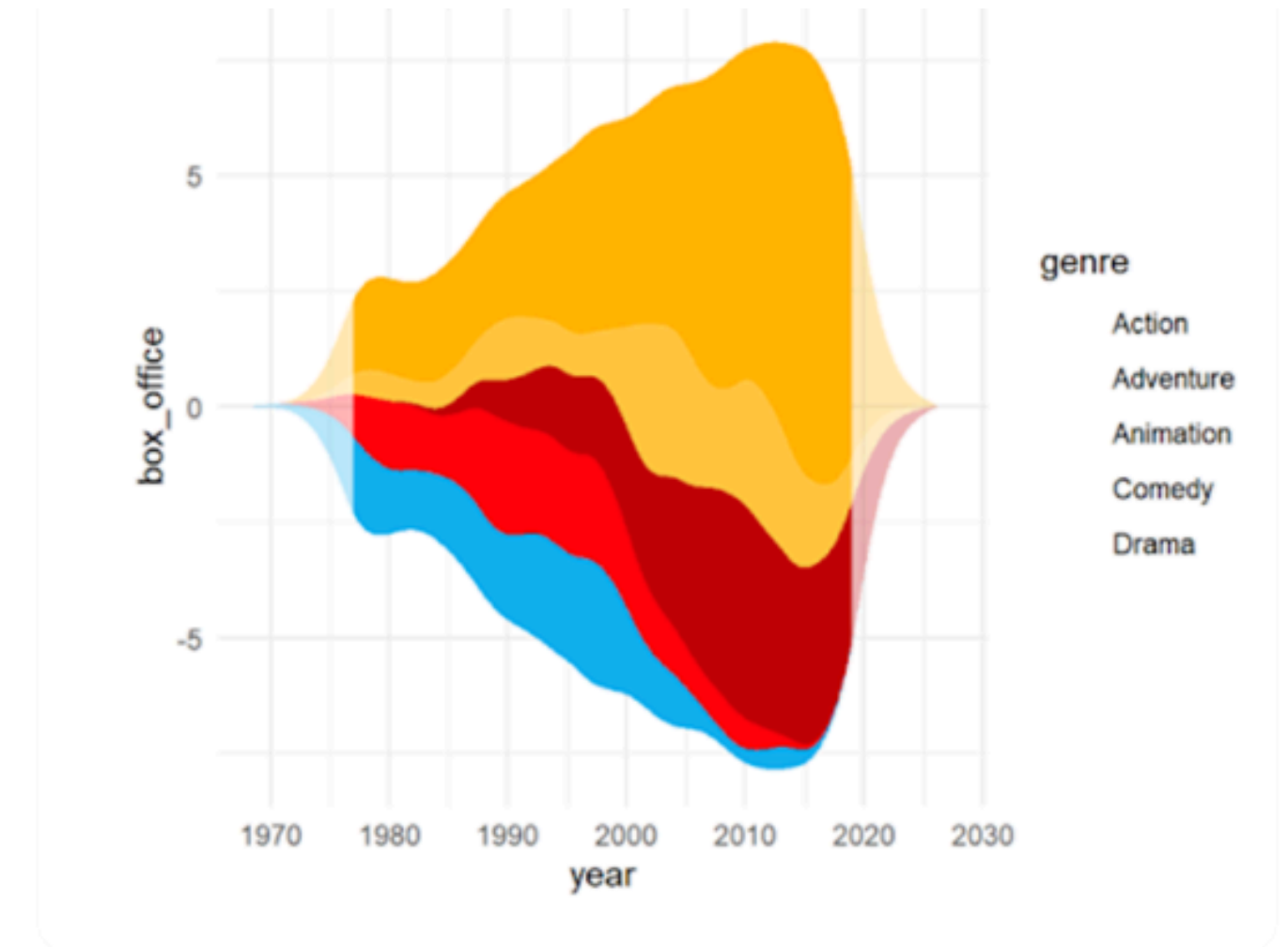
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6. Missing or Ambiguous Legend

- Detects the absence of a legend, incomplete or unclear labeling.



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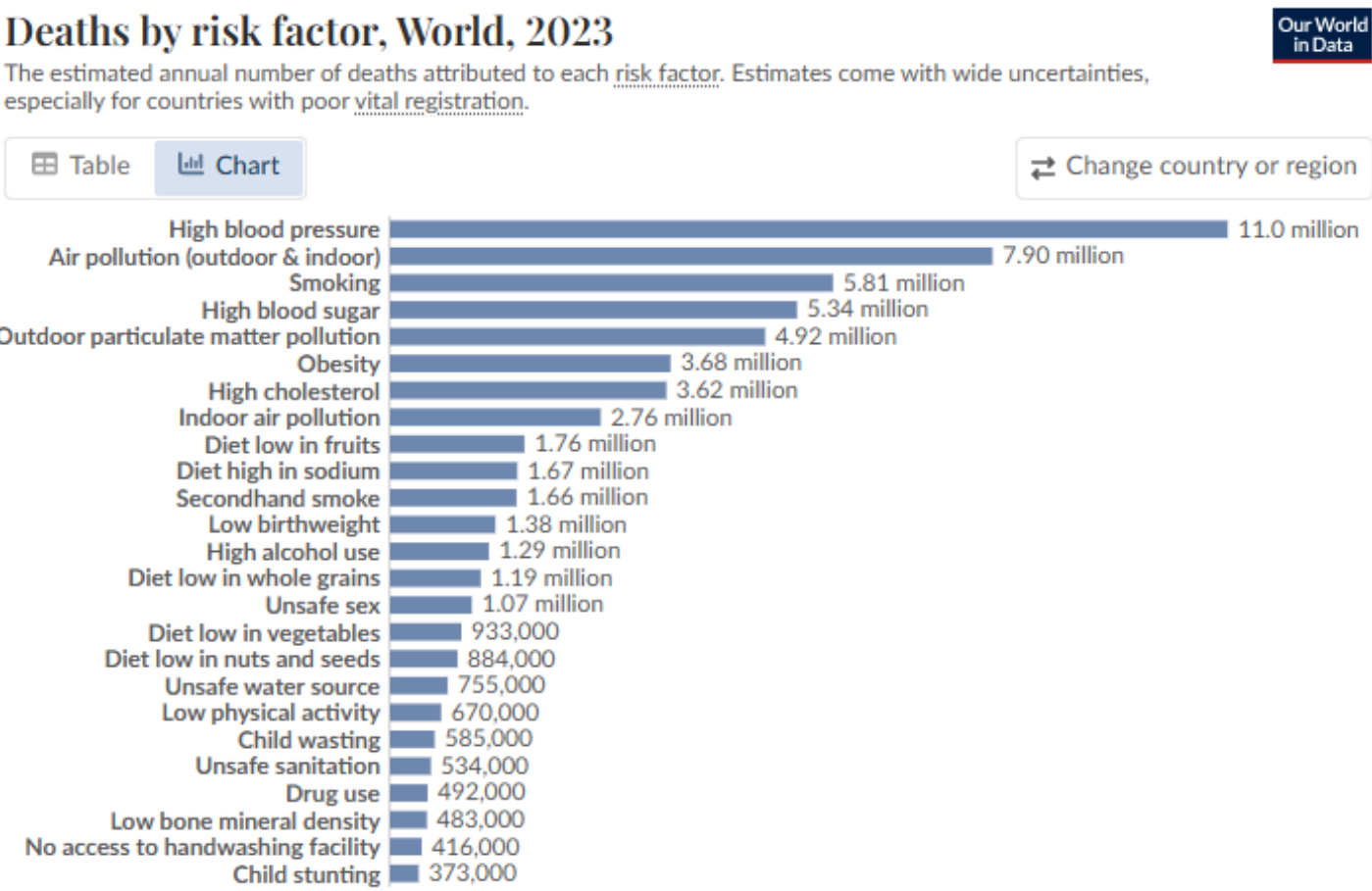
Evaluation Metrics

- Response Consistency
- Response Evaluation

Evaluation Metrics

Response Consistency

- Majority Voting Mechanism: Selecting the most frequent response
- Mitigate randomness and hallucinations



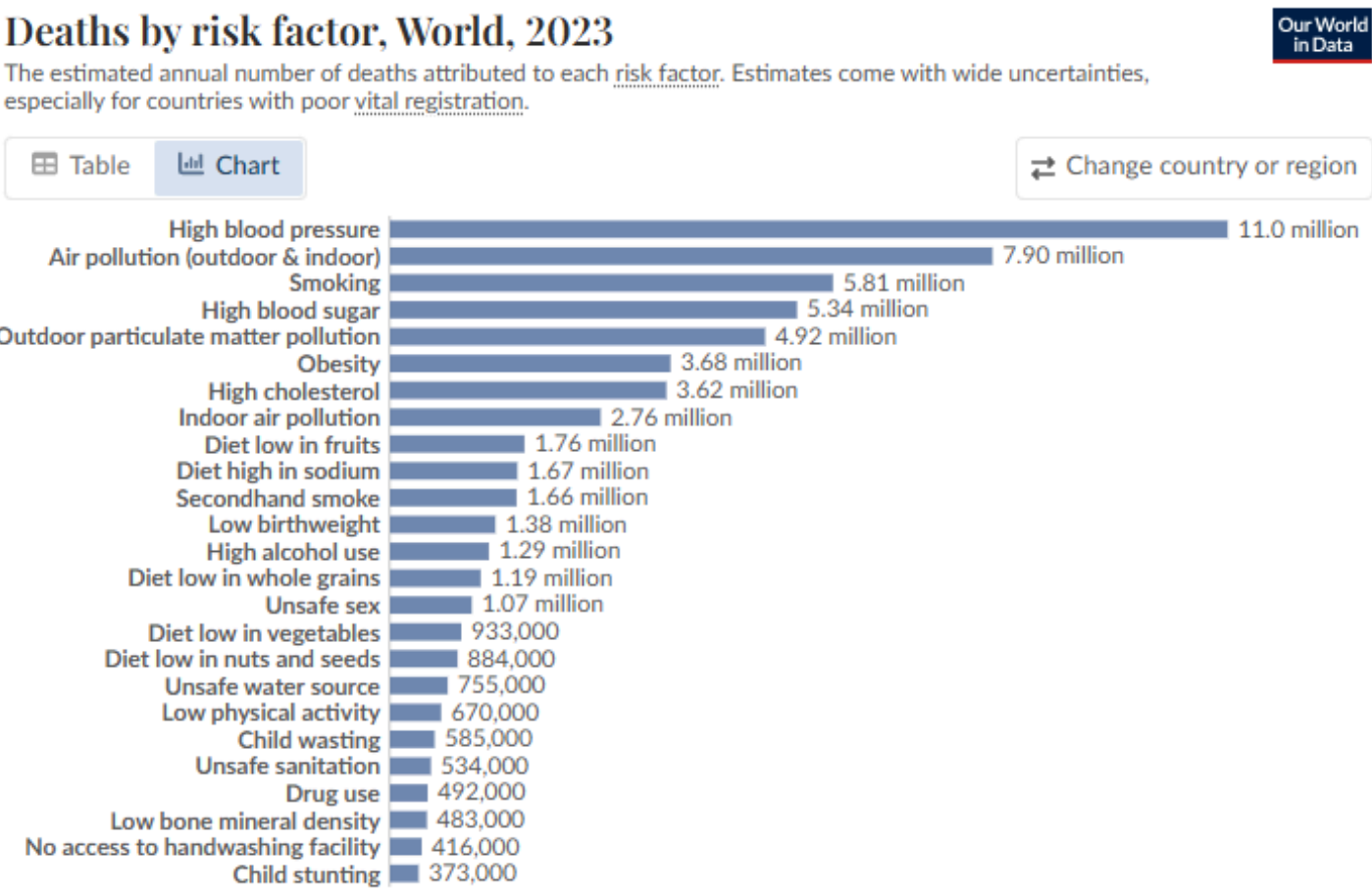
Question: How many risk factors result in deaths more than 5 million?				
Expected Output: 4				
Gemini (Trial 1)	Gemini (Trial 2)	Gemini (Trial 3)	Gemini (Trial 4)	Gemini (Trial 5)
There are 4 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 4 risk factors that result in more than 5 million deaths.

Evaluation Metrics

Response Consistency

- Majority Voting Mechanism: Selecting the most frequent response
- Mitigate randomness and hallucinations

Response Collected = 5



Question: How many risk factors result in deaths more than 5 million?

Expected Output: 4

Gemini (Trial 1)	Gemini (Trial 2)	Gemini (Trial 3)	Gemini (Trial 4)	Gemini (Trial 5)
There are 4 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 5 risk factors that result in more than 5 million deaths.	There are 4 risk factors that result in more than 5 million deaths.

Evaluation Metrics

Response Evaluation

Correct

Responses that are fully correct for each task, match the expected output and reflects the model's ability to produce reliable and precise outputs

Partially Correct

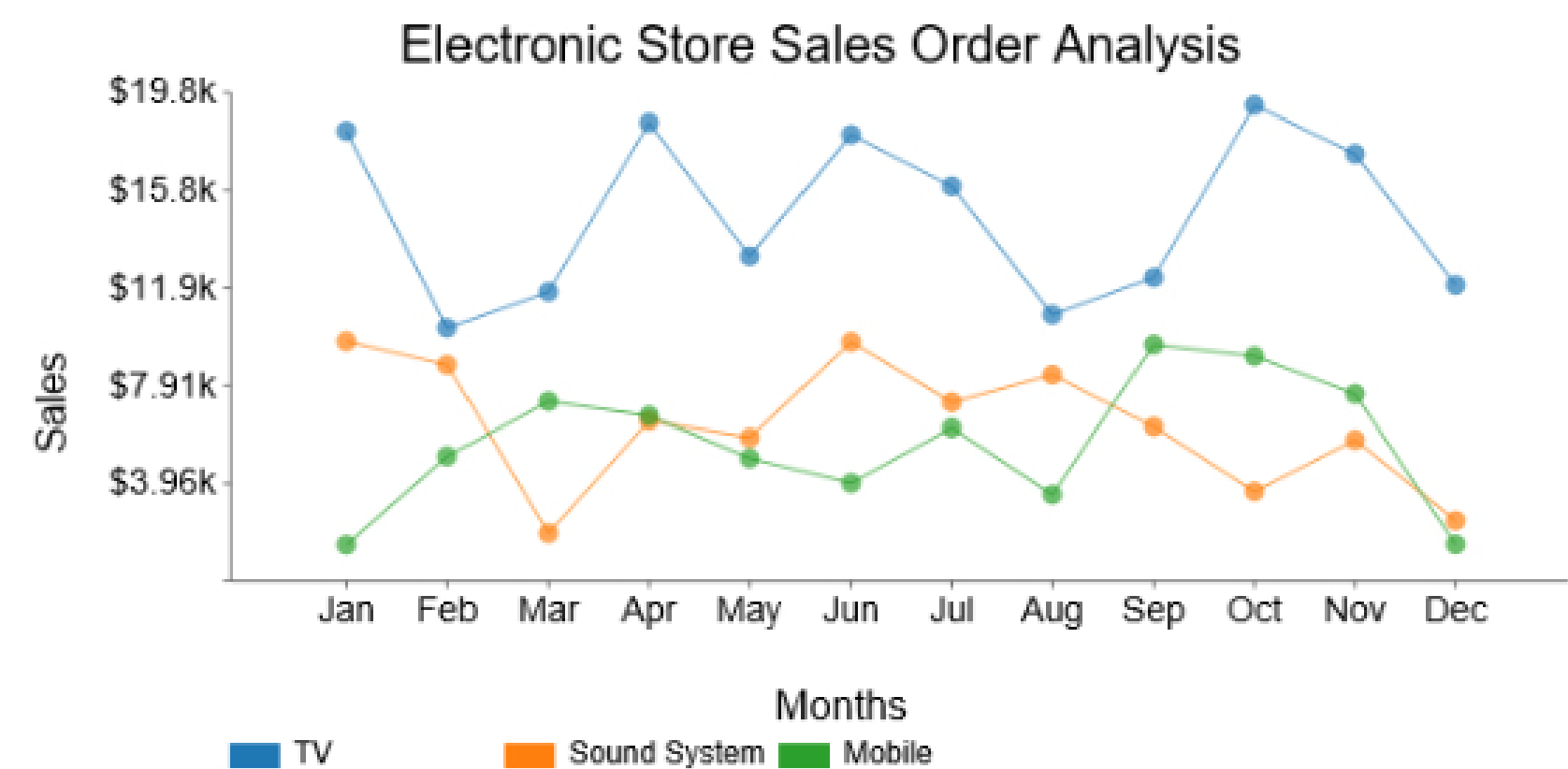
Captures responses that are generally correct but contain minor errors or the output is very close to the expected output.

Incorrect

Responses that are not aligned with the expected output and contain significant errors in interpretation or reasoning.

Evaluation Metrics

Task 1: Response Evaluation - Correct

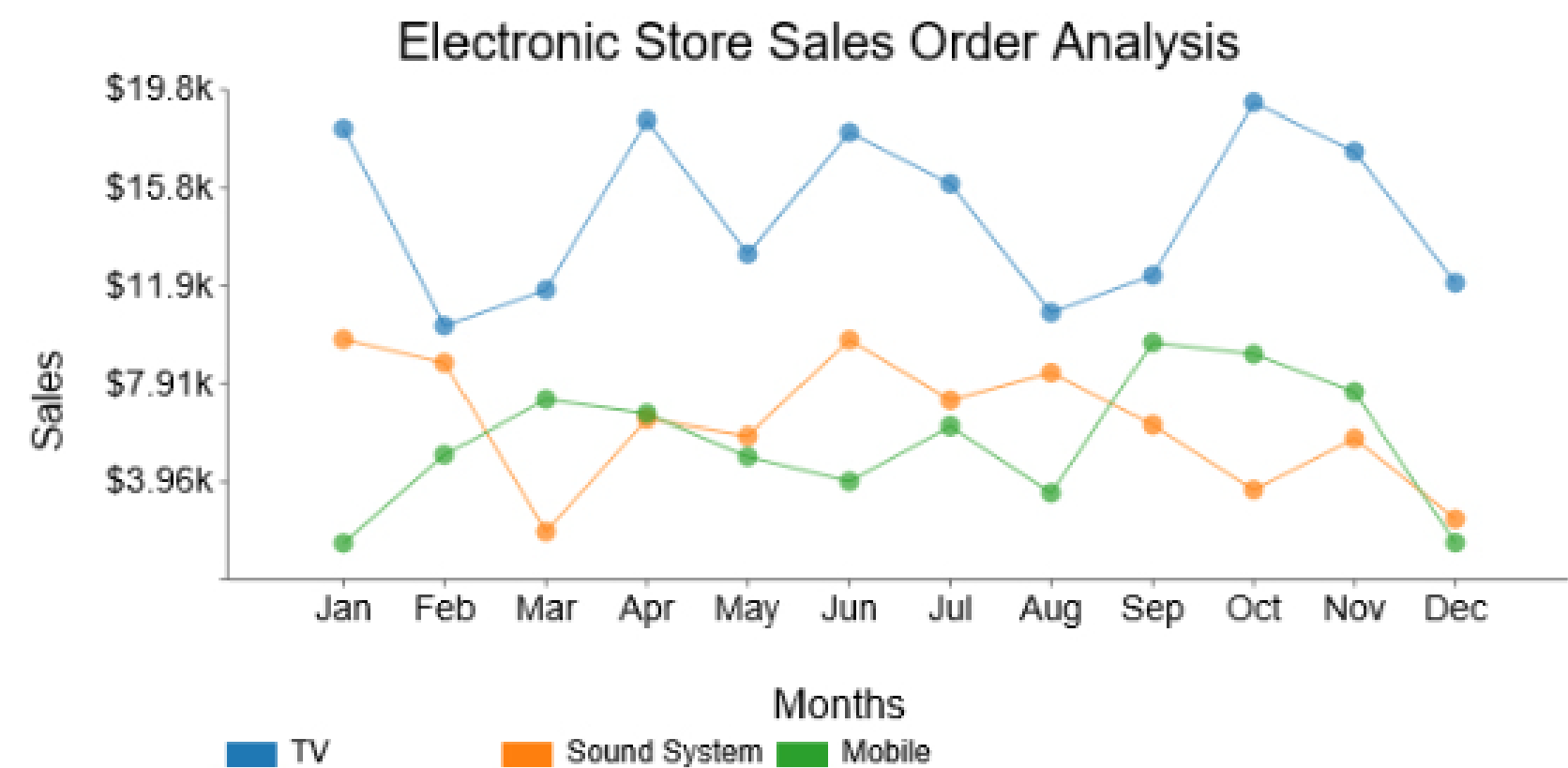


Example of Line Chart used for Task 1 (Visual Encoding Interpretation).

Which color represents the sales trend for TVs?			
Expected Output: Blue			
ChatGPT	Gemini	Claude	Mistral
The sales trend for TVs is represented by the blue line.	Blue represents the sales trend for TVs.	The blue line represents the sales trend for TVs.	The blue line in the chart represents the sales trend for TVs.

Evaluation Metrics

Task 1: Response Evaluation - Correct



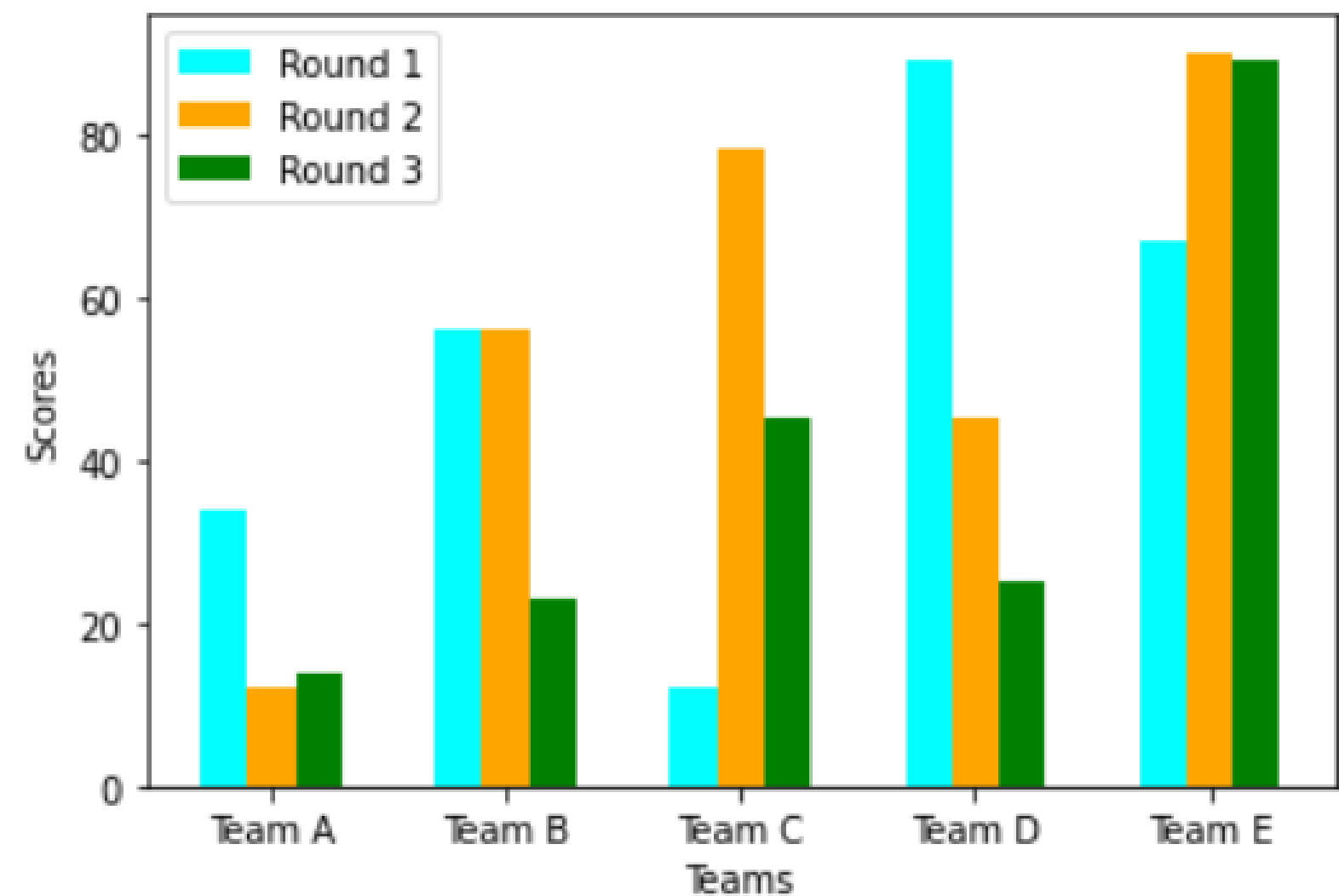
Example of Line Chart used for Task 1 (Visual Encoding Interpretation).

Which color represents the sales trend for TVs?			
Expected Output: Blue			
ChatGPT	Gemini	Claude	Mistral
The sales trend for TVs is represented by the blue line.	Blue represents the sales trend for TVs.	The blue line represents the sales trend for TVs.	The blue line in the chart represents the sales trend for TVs.

All responses are correct!

Evaluation Metrics

Task 2: Response Evaluation - Correct & Incorrect

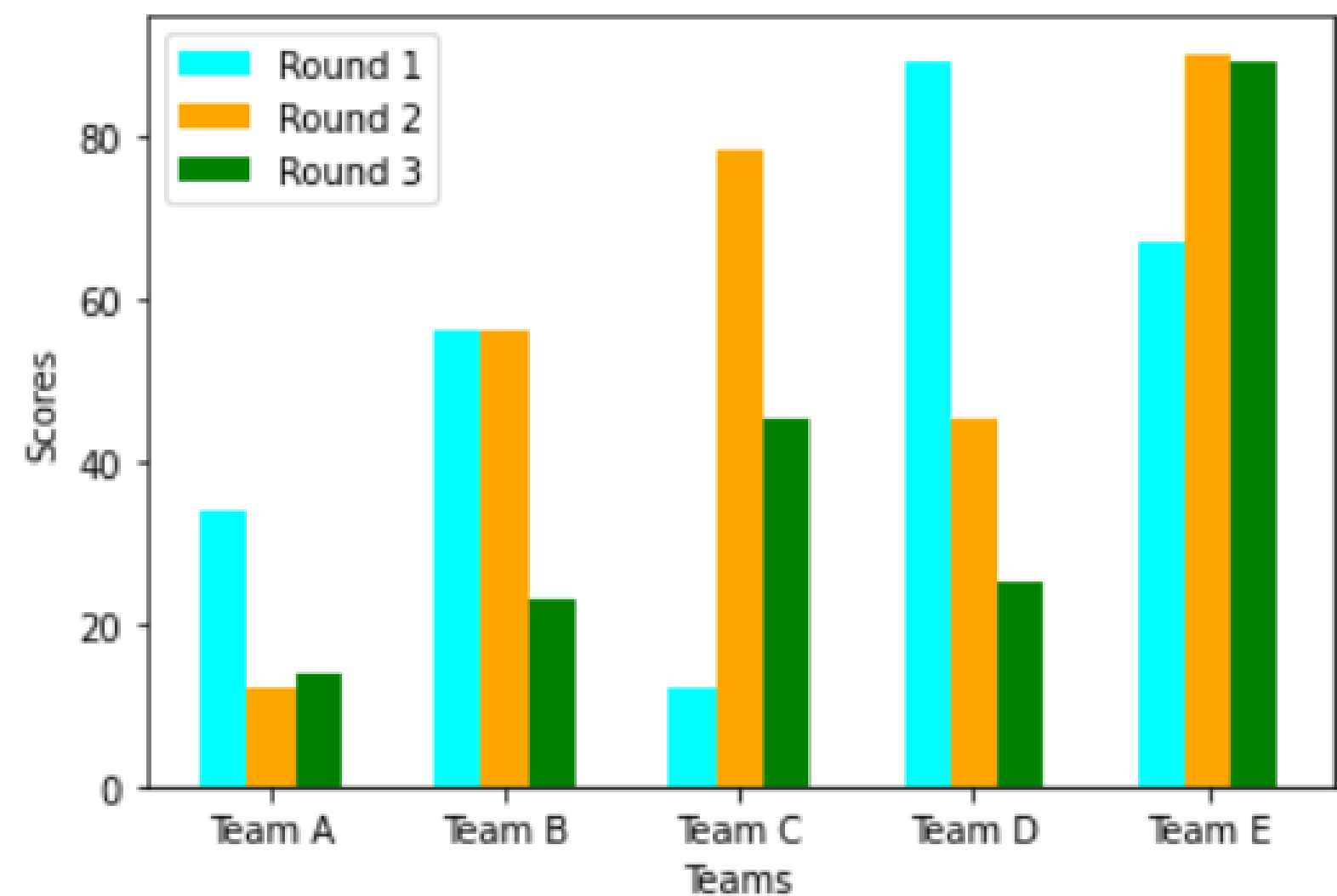


Example of Highest Value Identification question for Task2 (Chart-Based Question Answering) alongside responses generated by LLMs.

Which team has the highest score in Round 1?			
Expected Output: Team D			
ChatGPT	Gemini	Claude	Mistral
From the chart, Team D has the highest score in Round 1.	Team D has the highest score in Round 1.	Team D has the highest score in Round 1, with the cyan bar reaching approximately 88 points.	The bar chart shows that Team E has the highest score in Round 1, with a score of around 80.

Evaluation Metrics

Task 2: Response Evaluation - Correct & Incorrect



Example of Highest Value Identification question for Task2 (Chart-Based Question Answering) alongside responses generated by LLMs.

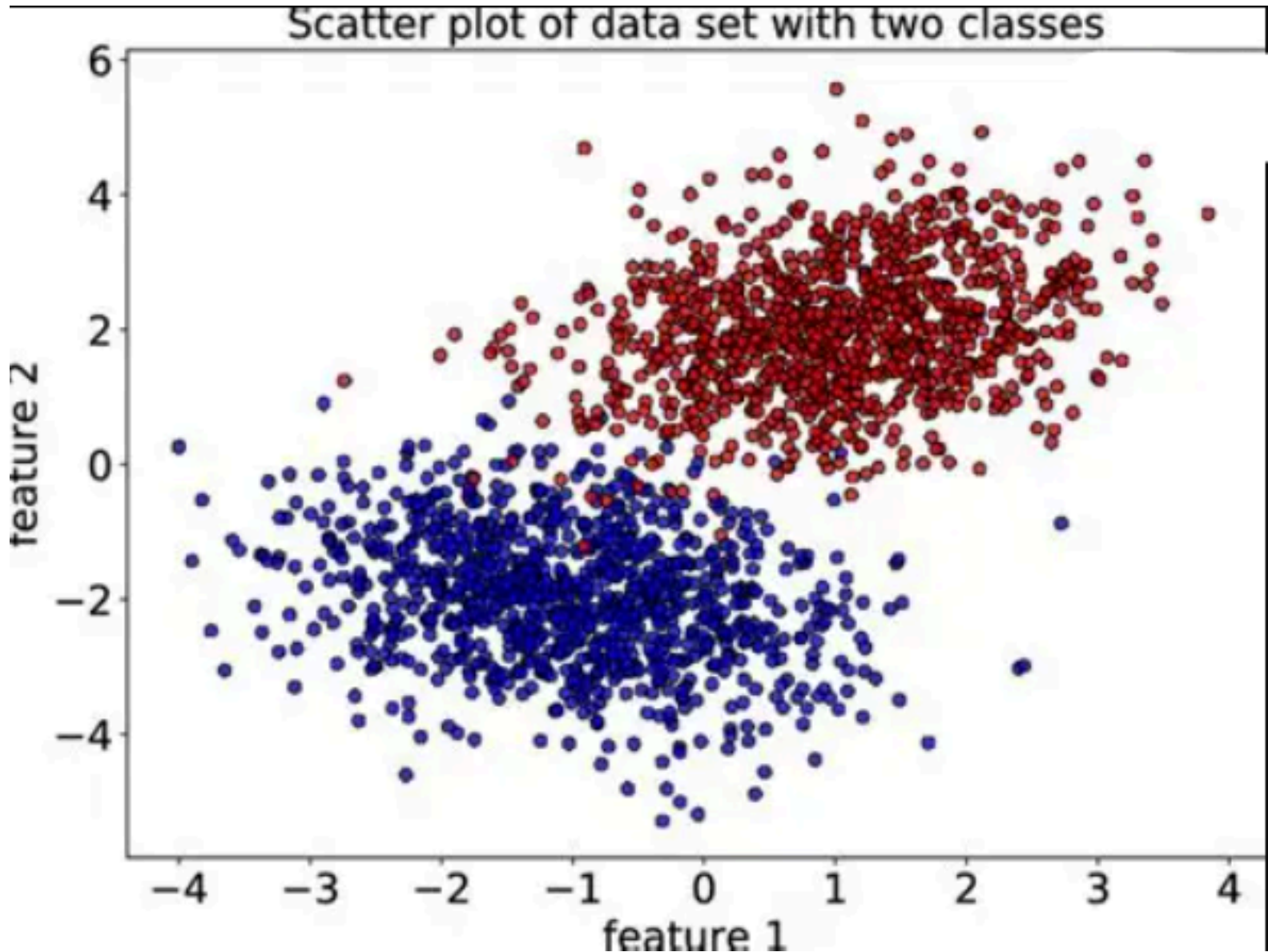
Which team has the highest score in Round 1?			
Expected Output: Team D			
ChatGPT	Gemini	Claude	Mistral
From the chart, Team D has the highest score in Round 1.	Team D has the highest score in Round 1.	Team D has the highest score in Round 1, with the cyan bar reaching approximately 88 points.	The bar chart shows that Team E has the highest score in Round 1, with a score of around 80.

Correct

Incorrect

Evaluation Metrics

Task 3: Response Evaluation - Correct

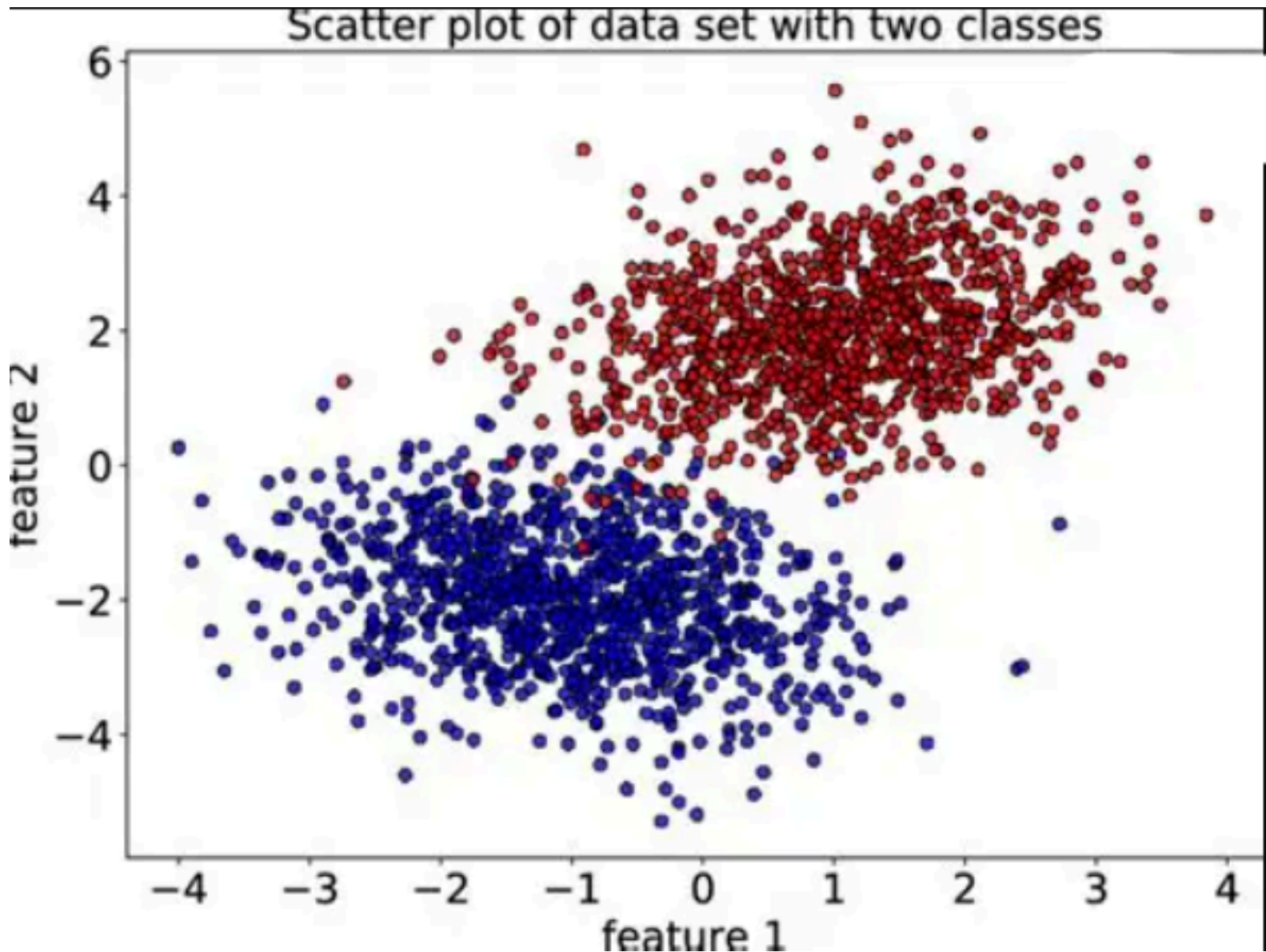


Example of scatter plot with missing legend used for Task 3 (Detection of Misleading Charts).

Are there any flaws in the chart? Detect what is wrong and explain why.
Expected Output: Missing Legend
ChatGPT • Missing legend: Red and blue classes are not labeled. • Poor class labeling: Title says “two classes” but classes are unnamed. • Overplotting: Many overlapping points make density unclear. • Limited context: Axes labeled generically as “feature 1” and “feature 2”.
Gemini • Missing legend: No explanation of colors. • Overplotting: Dense clusters obscure individual points. • Lack of units: Axes generic with no physical meaning. • Class overlap: Red and blue points overlap significantly. • Poor margin usage: Points close to borders.
Claude • Missing legend: Classes not labeled. • Overlapping points: Dense regions hard to interpret. • Generic title: "Scatter plot of data set with two classes". • No class labels anywhere. • Generic axis labels: Feature names are placeholders. • No gridlines: Hard to estimate point positions.
Mistral • Overlapping points obscure distribution. • Lack of transparency: Points fully opaque. • No clear separation between classes. • Generic axes and title. • Missing legend: Red/blue colors not explained. • Suggested improvements: Use transparency, add legend, provide context, consider alternative visualizations, add gridlines.

Evaluation Metrics

Task 3: Response Evaluation - Correct



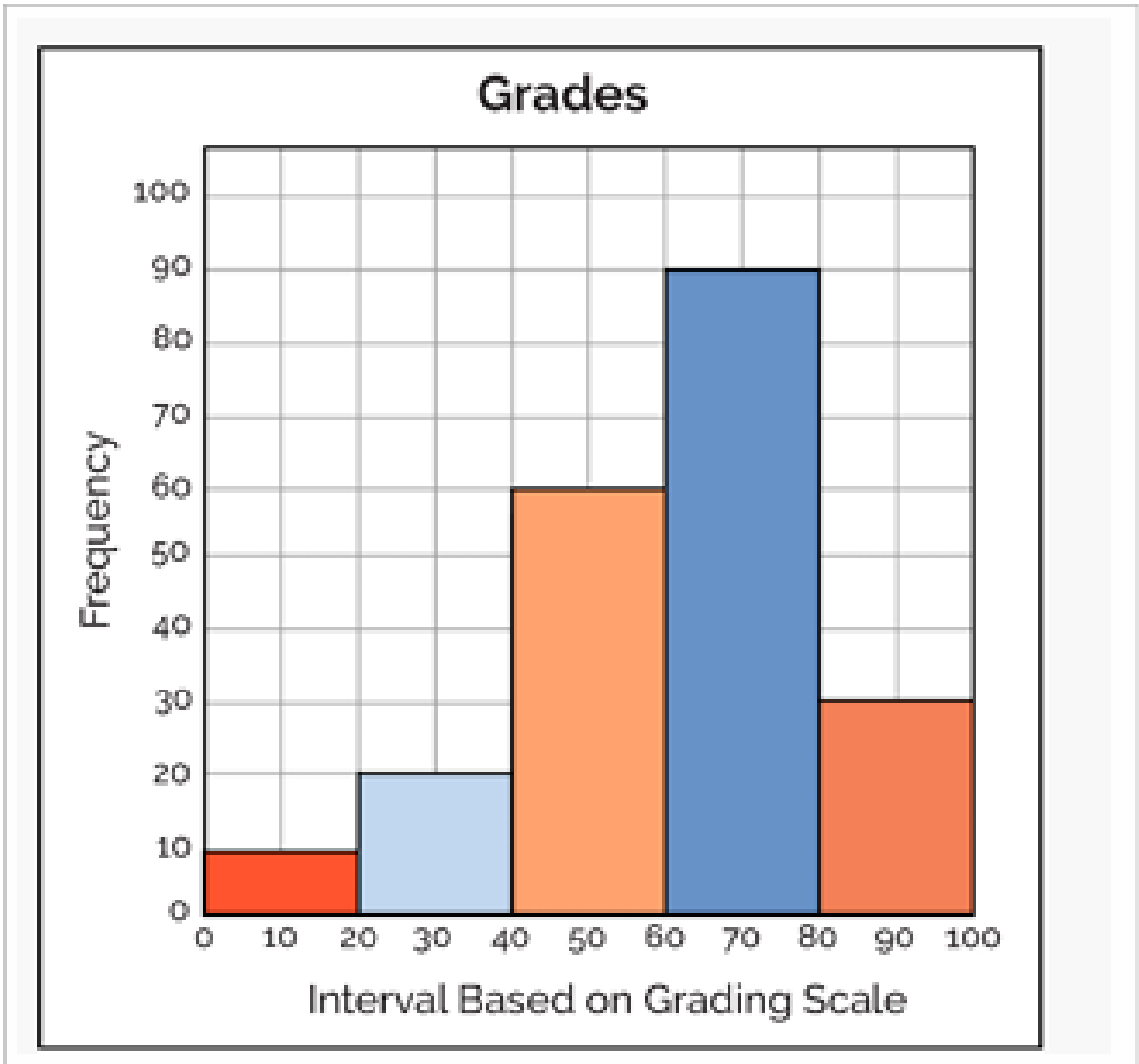
All responses are correct!

Example of scatter plot with missing legend used for Task 3 (Detection of Misleading Charts).

Are there any flaws in the chart? Detect what is wrong and explain why.
Expected Output: Missing Legend
ChatGPT • Missing legend: Red and blue classes are not labeled. • Poor class labeling: Title says “two classes” but classes are unnamed. • Overplotting: Many overlapping points make density unclear. • Limited context: Axes labeled generically as “feature 1” and “feature 2”.
Gemini • Missing legend: No explanation of colors. • Overplotting: Dense clusters obscure individual points. • Lack of units: Axes generic with no physical meaning. • Class overlap: Red and blue points overlap significantly. • Poor margin usage: Points close to borders.
Claude • Missing legend: Classes not labeled. • Overlapping points: Dense regions hard to interpret. • Generic title: "Scatter plot of data set with two classes". • No class labels anywhere. • Generic axis labels: Feature names are placeholders. • No gridlines: Hard to estimate point positions.
Mistral • Overlapping points obscure distribution. • Lack of transparency: Points fully opaque. • No clear separation between classes. • Generic axes and title. • Missing legend: Red/blue colors not explained. • Suggested improvements: Use transparency, add legend, provide context, consider alternative visualizations, add gridlines.

Evaluation Metrics

Task 1: Response Evaluation - Partially Correct

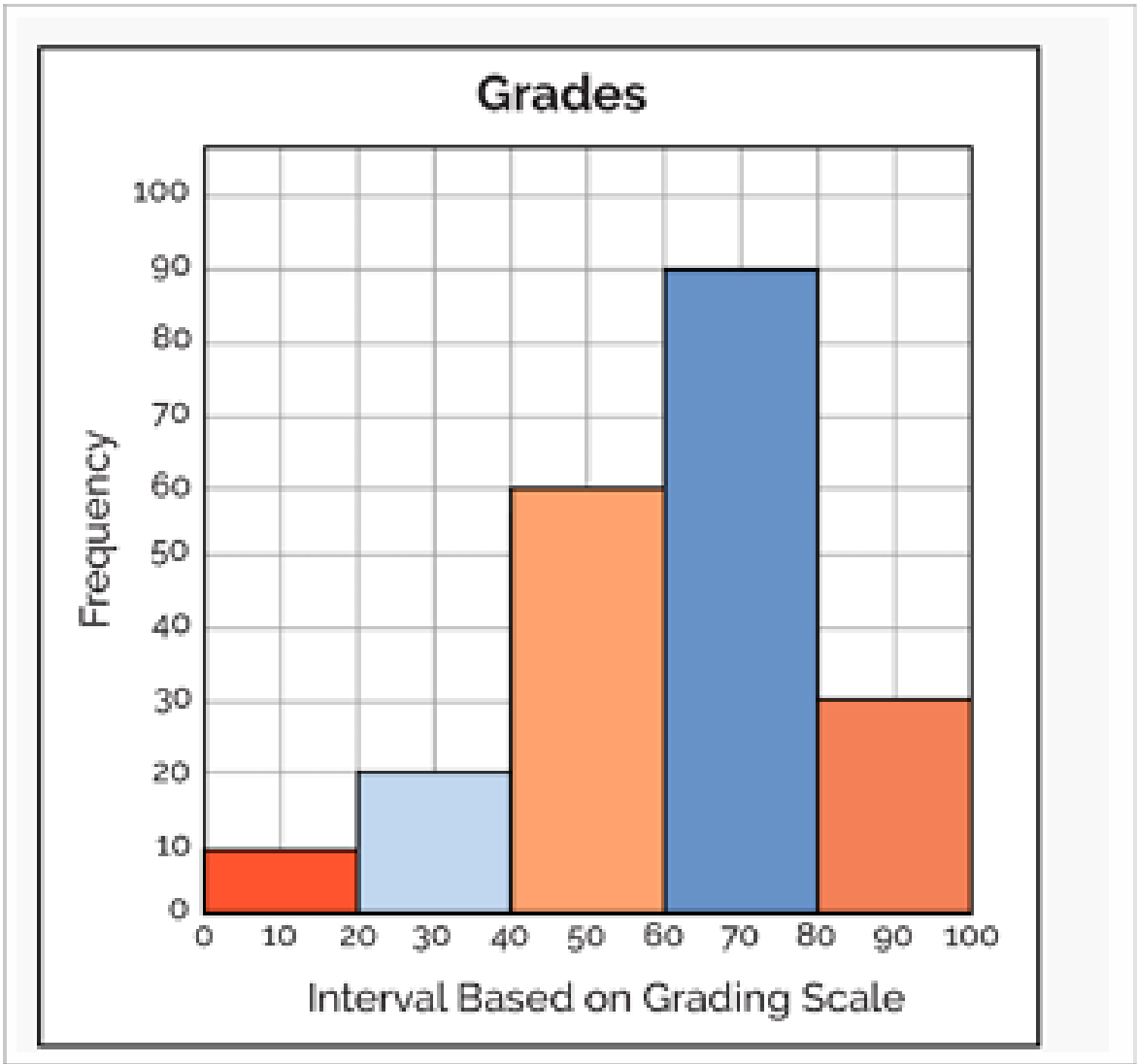


Example of histogram used for Task 1 (Visual Encoding Interpretation) provided to ChatGPT.

Question: Which color represents the bar for the lowest grade interval (0–20)?			
Expected Output: Dark Orange			
ChatGPT	Gemini	Claude	Mistral
The bar for the lowest grade interval 0–20 is the reddish-orange (bright red/orange) bar on the far left.	Based on the histogram provided, the bar for the lowest grade interval (0–20) is represented by the color orange (specifically a bright, reddish-orange).	The red/orange-red color represents the bar for the lowest grade interval (0–10), which has a frequency of 10.	In the histogram, the red bar represents the lowest grade interval (0–20).

Evaluation Metrics

Task 1: Response Evaluation - Partially Correct



Example of histogram used for Task 1 (Visual Encoding Interpretation) provided to ChatGPT.

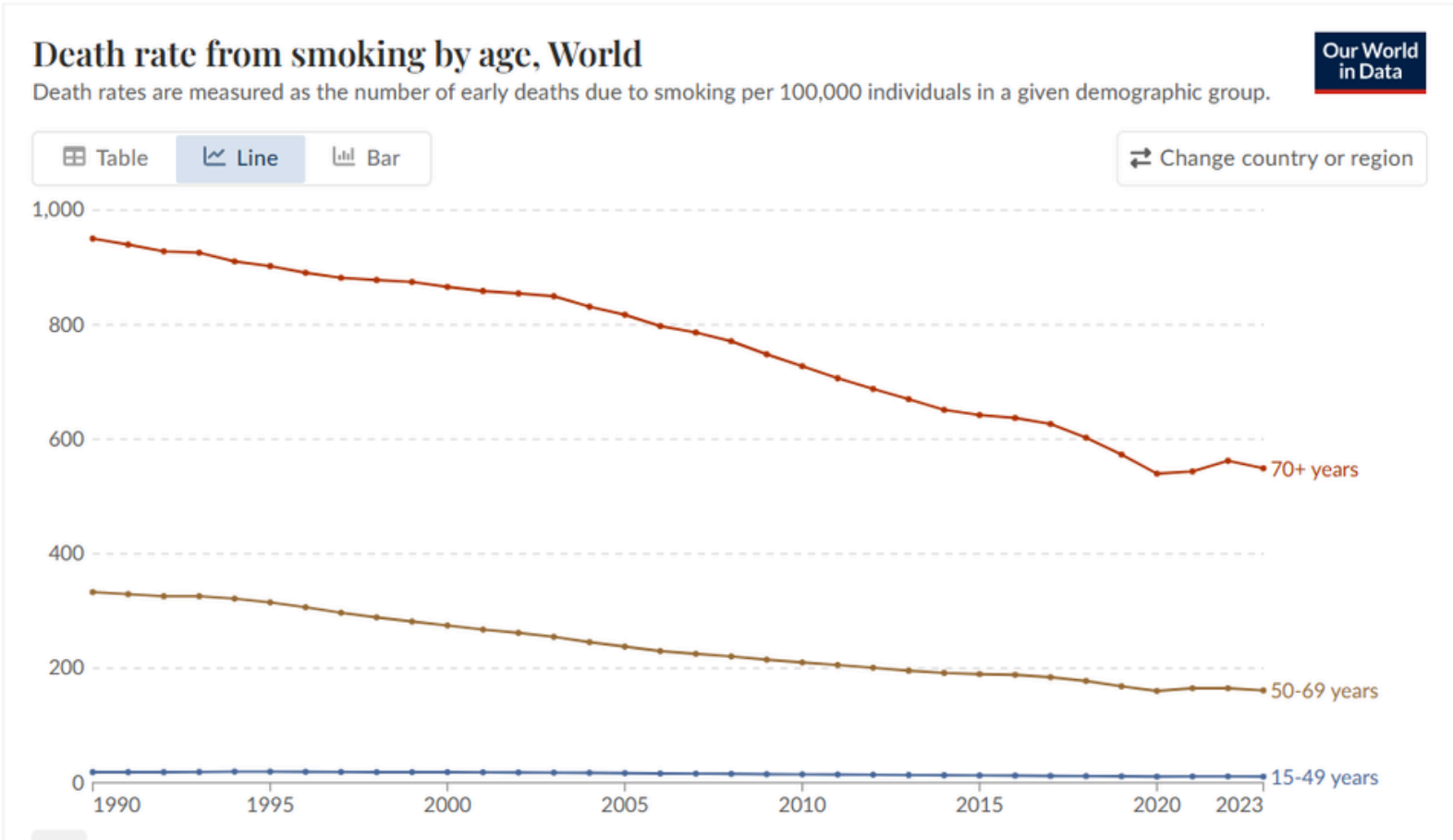
Question: Which color represents the bar for the lowest grade interval (0–20)?			
Expected Output: Dark Orange			
ChatGPT	Gemini	Claude	Mistral
The bar for the lowest grade interval 0–20 is the reddish-orange (bright red/orange) bar on the far left.	Based on the histogram provided, the bar for the lowest grade interval (0–20) is represented by the color orange (specifically a bright, reddish-orange).	The red/orange-red color represents the bar for the lowest grade interval (0–10), which has a frequency of 10.	In the histogram, the red bar represents the lowest grade interval (0–20).

Correct

Partially Correct

Evaluation Metrics

Task 2: Response Evaluation - Partially Correct

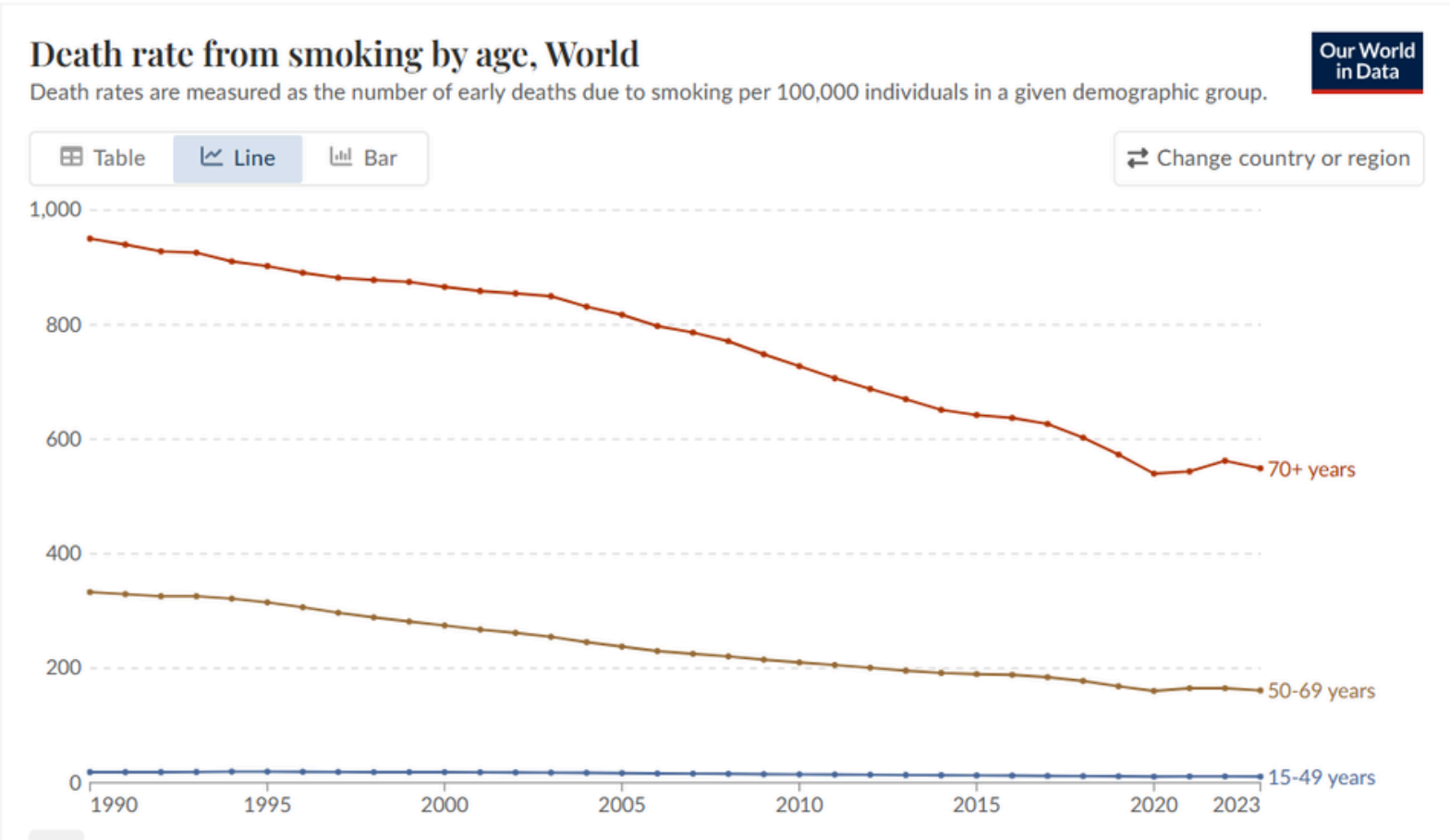


Example of Line chart used for Task 2 (Chart-Based Question Answering)

What is the approximate value for the 50–69 years group in 2021?			
Expected Output: 165			
ChatGPT	Gemini	Claude	Mistral
From the chart, the 50–69 years line in 2021 sits just above the 150 mark and below 170.	Based on the chart, the approximate value for the 50–69 years group in 2021 is 165.	The approximate value for the 50–69 years group in 2021 is around 175 per 100,000 individuals.	The approximate death rate from smoking for the 50–69 years age group in 2021 is around 180 per 100,000 individuals.

Evaluation Metrics

Task 2: Response Evaluation - Partially Correct & Incorrect



Example of Line chart used for Task 2 (Chart-Based Question Answering)

What is the approximate value for the 50–69 years group in 2021?			
Expected Output: 165			
ChatGPT	Gemini	Claude	Mistral
From the chart, the 50–69 years line in 2021 sits just above the 150 mark and below 170.	Based on the chart, the approximate value for the 50–69 years group in 2021 is 165.	The approximate value for the 50–69 years group in 2021 is around 175 per 100,000 individuals.	The approximate death rate from smoking for the 50–69 years age group in 2021 is around 180 per 100,000 individuals.

↓
Partially Correct

↓
Correct

↘
Incorrect

↘
Incorrect

Results

Results: Task 1

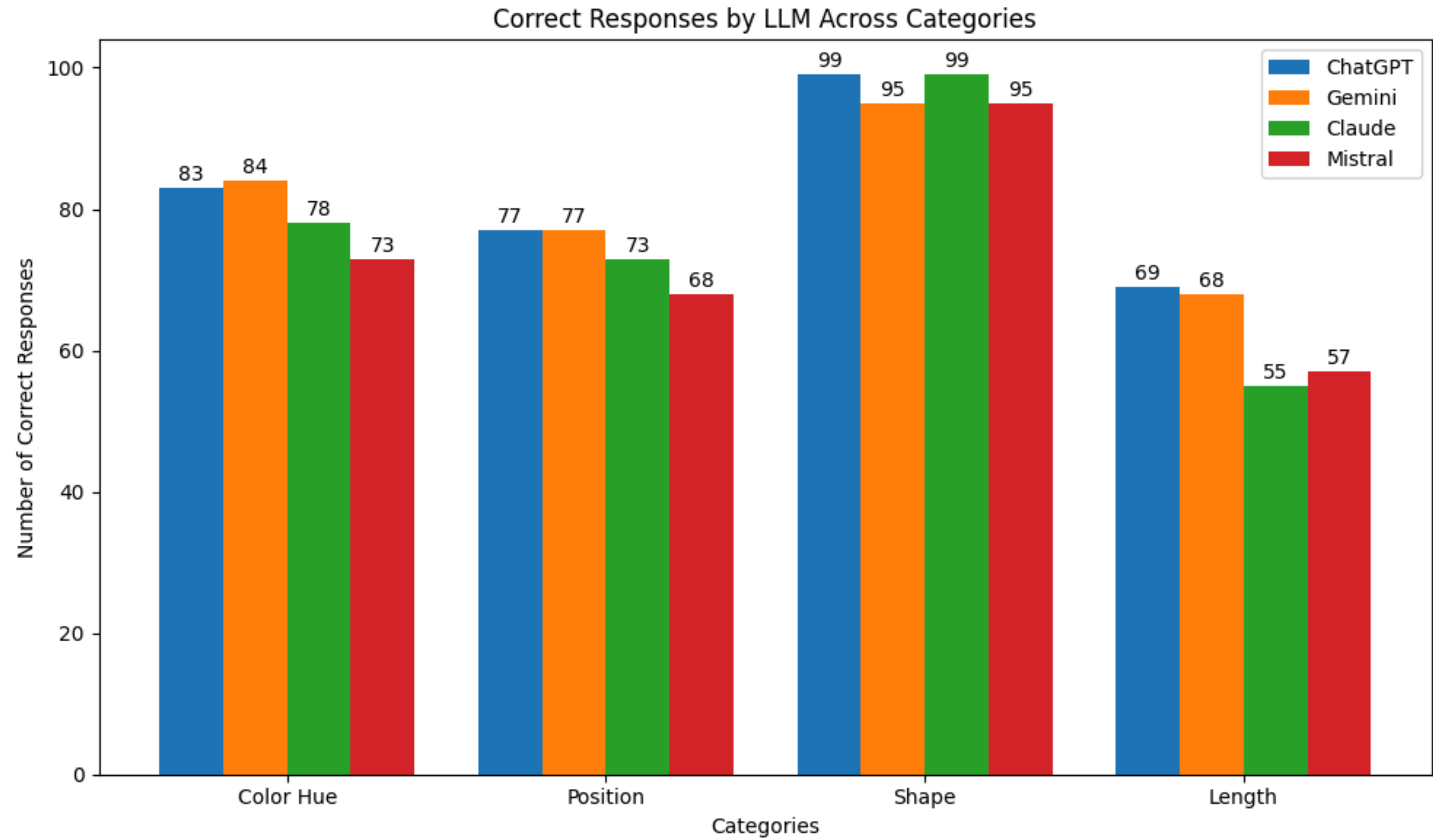
Chart No.	Chart Category	No. of Chart Types	No. of Categories	Questions per Category	Total Questions
1	Common	10	4	5	200
2	Advanced	10	4	5	200
Total					400

LLMs	Correct Responses	Partially Correct Responses	Incorrect Responses
ChatGPT	328	18	54
Gemini	324	19	57
Claude	305	26	69
Mistral	293	18	89

Results: Task 1

Category	Evaluation	ChatGPT	Gemini	Claude	Mistral
Color Hue	Correct	83	84	78	73
	Partially Correct	6	5	12	6
	Incorrect	11	11	10	21
Position	Correct	77	77	73	68
	Partially Correct	6	4	6	6
	Incorrect	17	19	21	26
Shape	Correct	99	95	99	95
	Partially Correct	0	3	0	1
	Incorrect	1	2	1	4
Length (Height/Size)	Correct	69	68	55	57
	Partially Correct	6	7	8	5
	Incorrect	25	25	37	38

Results: Task 1



Results: Task 2

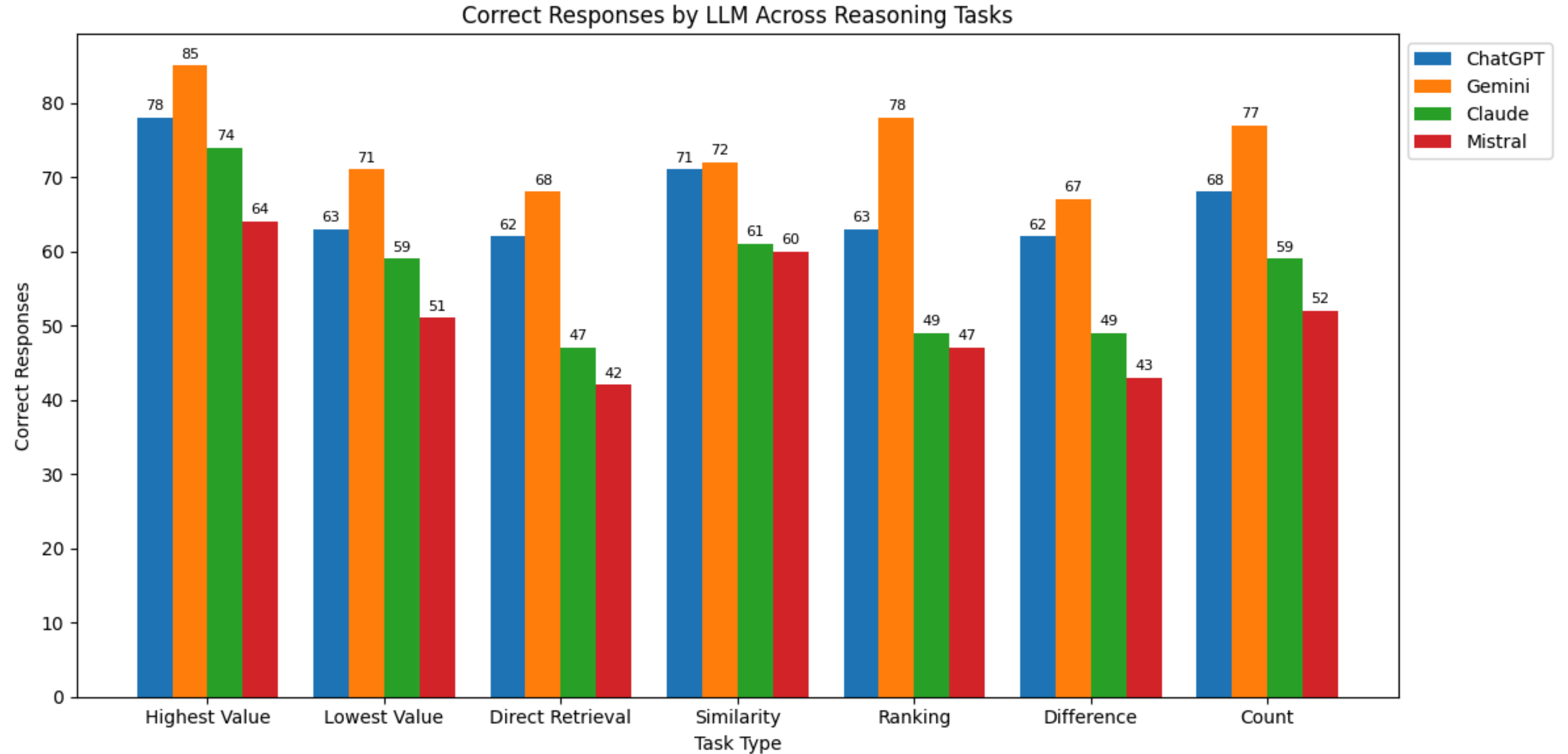
Chart No.	Chart Category	No. of Chart Types	Categories	Questions per Category	Total Questions
1	Common	10	7	5	350
2	Advanced	10	7	5	350
Total			14		700

LLMs	Correct Responses	Partially Correct Responses	Incorrect Responses
ChatGPT	467	40	193
Gemini	518	45	137
Claude	398	63	239
Mistral	359	42	299

Results: Task 2

Category	Evaluation	ChatGPT	Gemini	Claude	Mistral
Highest Value Identification	Correct	78	85	74	64
	Partially Correct	5	4	4	4
	Incorrect	17	11	22	32
Lowest Value Identification	Correct	63	71	59	51
	Partially Correct	6	8	11	7
	Incorrect	31	21	30	42
Direct Value Retrieval	Correct	62	68	47	42
	Partially Correct	6	9	15	7
	Incorrect	32	23	38	51
Similarity Detection	Correct	71	72	61	60
	Partially Correct	5	4	5	3
	Incorrect	24	24	34	37
Ranking	Correct	63	78	49	47
	Partially Correct	6	4	7	5
	Incorrect	31	18	44	48
Difference between Values	Correct	62	67	49	43
	Partially Correct	7	9	13	13
	Incorrect	31	24	38	44
Counting-Based Reasoning	Correct	68	77	59	52
	Partially Correct	5	7	8	3
	Incorrect	27	16	33	45

Results: Task 2



Results: Task 3

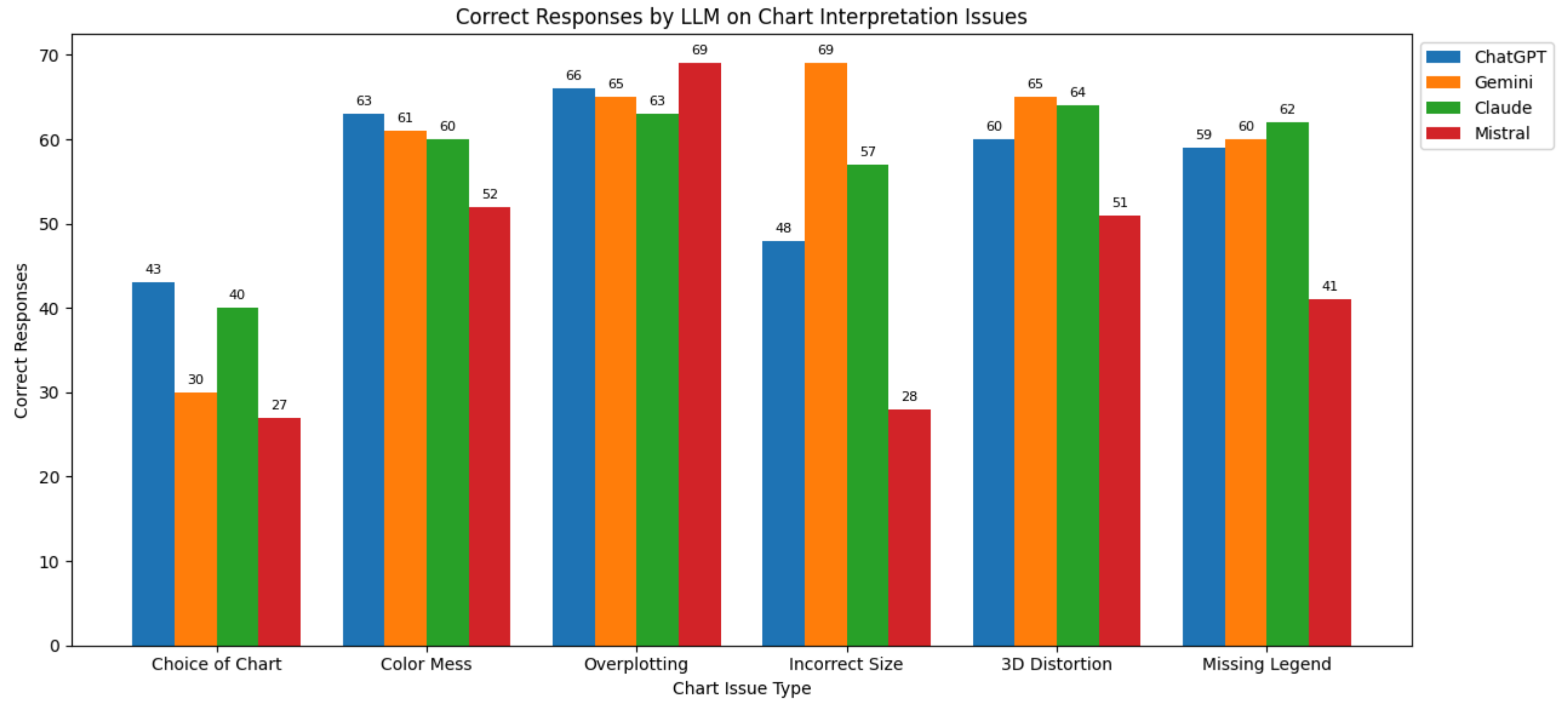
Chart No.	Chart Category	No. of Chart Types	Categories per Chart	Questions per Category	Total Questions
1	Common	10	6	4	240
2	Advanced	8	6	4 ¹	198
Total		18			438

LLMs	Correct Responses	Partially Correct Responses	Incorrect Responses
ChatGPT	339	19	80
Gemini	350	15	73
Claude	346	14	78
Mistral	268	10	160

Results: Task 3

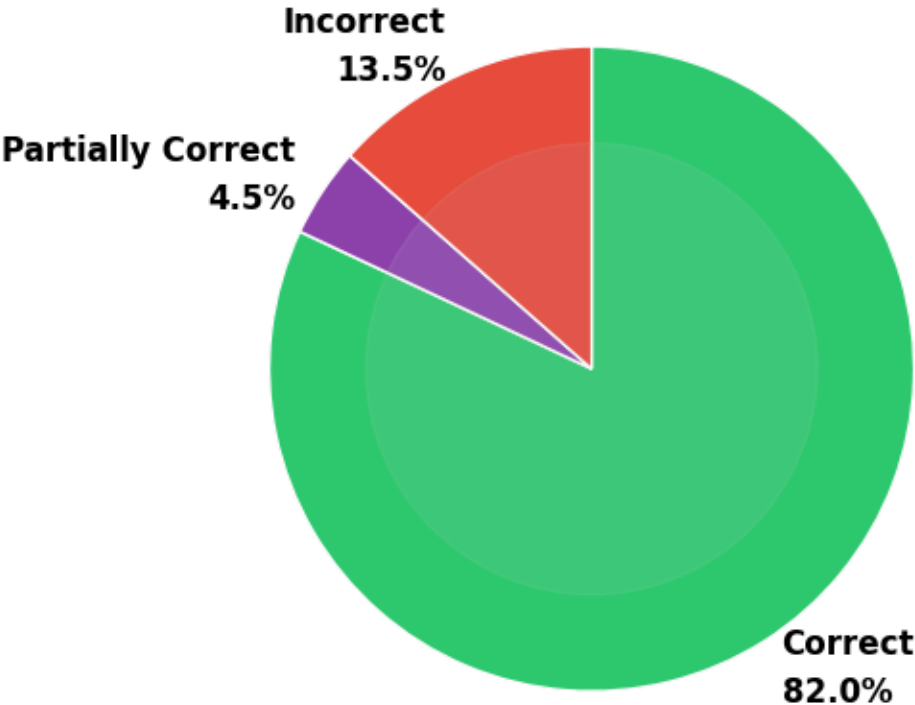
Category	Evaluation	ChatGPT	Gemini	Claude	Mistral
Choice of Chart	Correct	43	30	40	27
	Partially Correct	5	4	4	2
	Incorrect	25	39	29	44
Color Mess	Correct	63	61	60	52
	Partially Correct	2	1	2	3
	Incorrect	8	11	11	18
Overplotting	Correct	66	65	63	69
	Partially Correct	4	5	1	1
	Incorrect	3	3	9	3
Incorrect Size Encoding	Correct	48	69	57	28
	Partially Correct	0	1	4	1
	Incorrect	25	3	12	44
3D Distortion	Correct	60	65	64	51
	Partially Correct	4	1	1	0
	Incorrect	9	7	8	22
Missing or Ambiguous Legend	Correct	59	60	62	41
	Partially Correct	4	3	2	3
	Incorrect	10	10	9	29

Results: Task 3

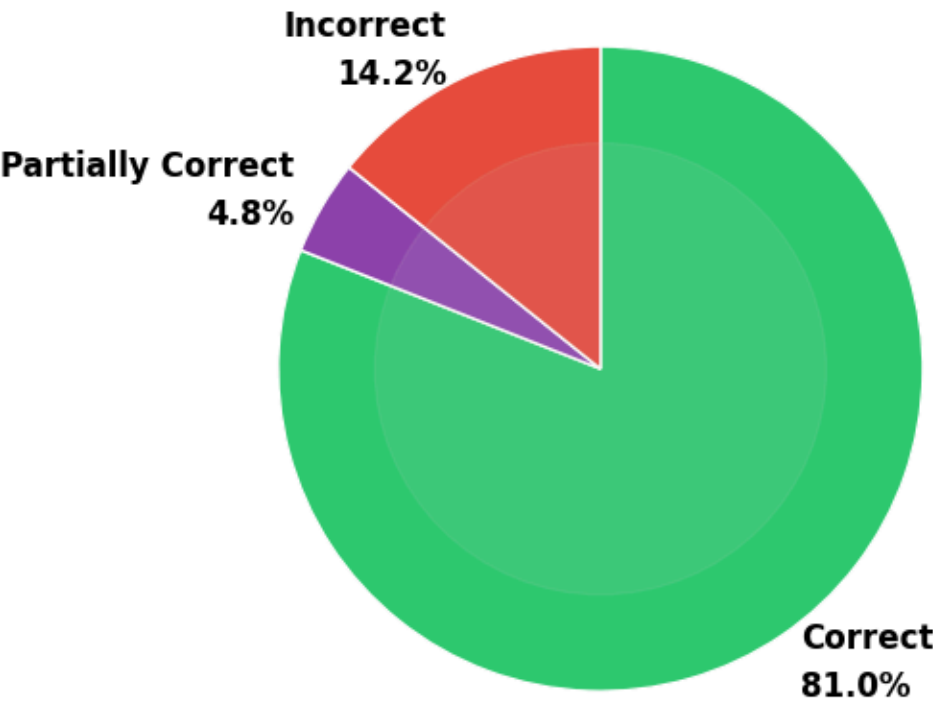


Error Analysis: Task 1

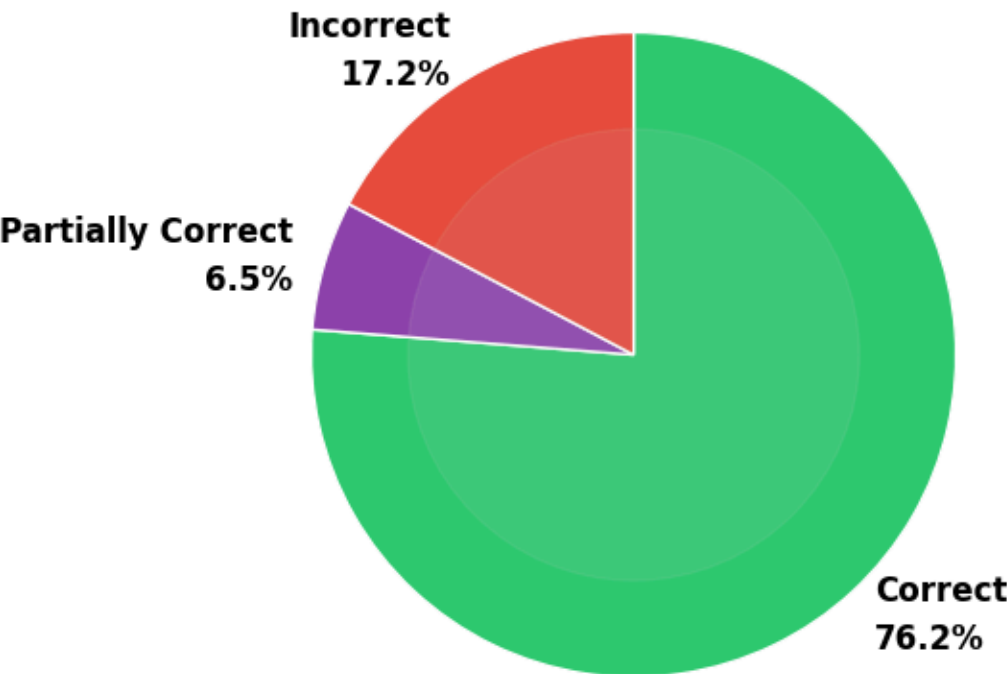
ChatGPT



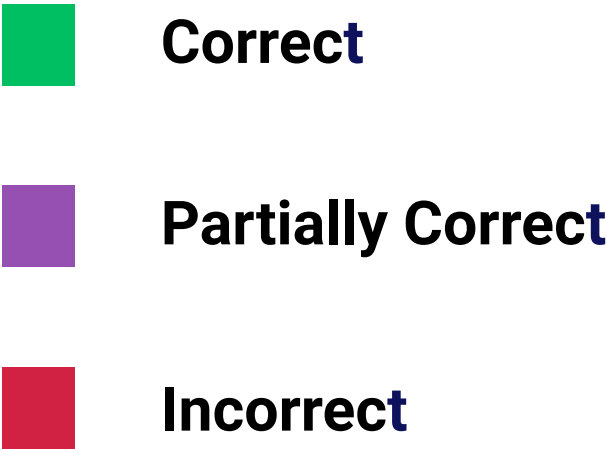
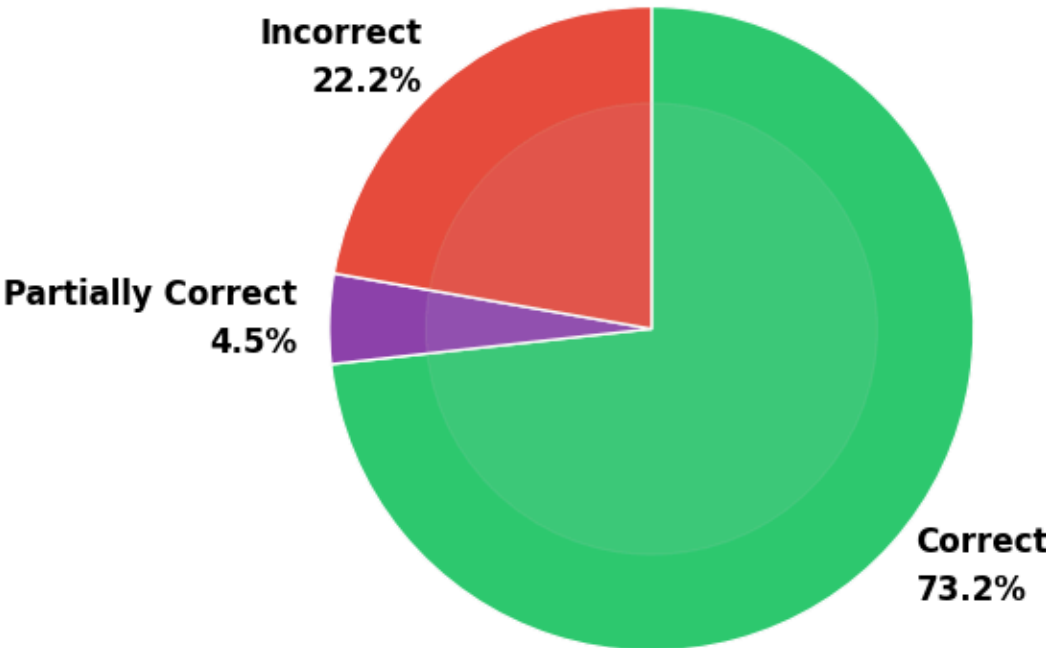
Gemini



Claude

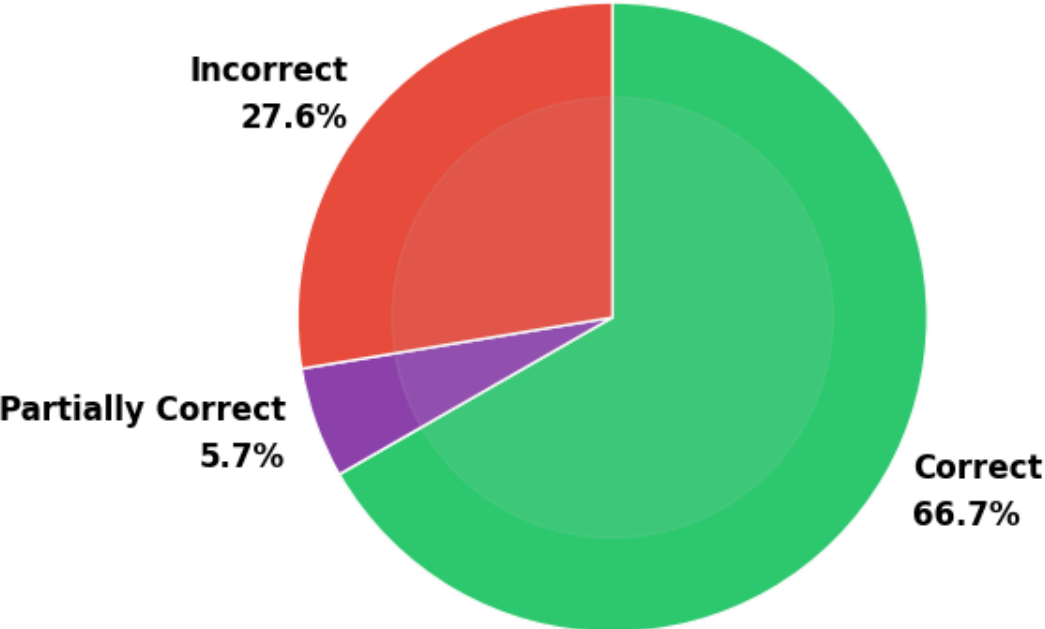


Mistral

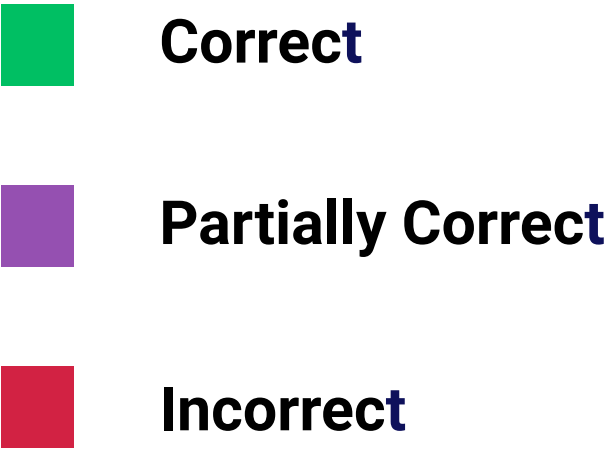
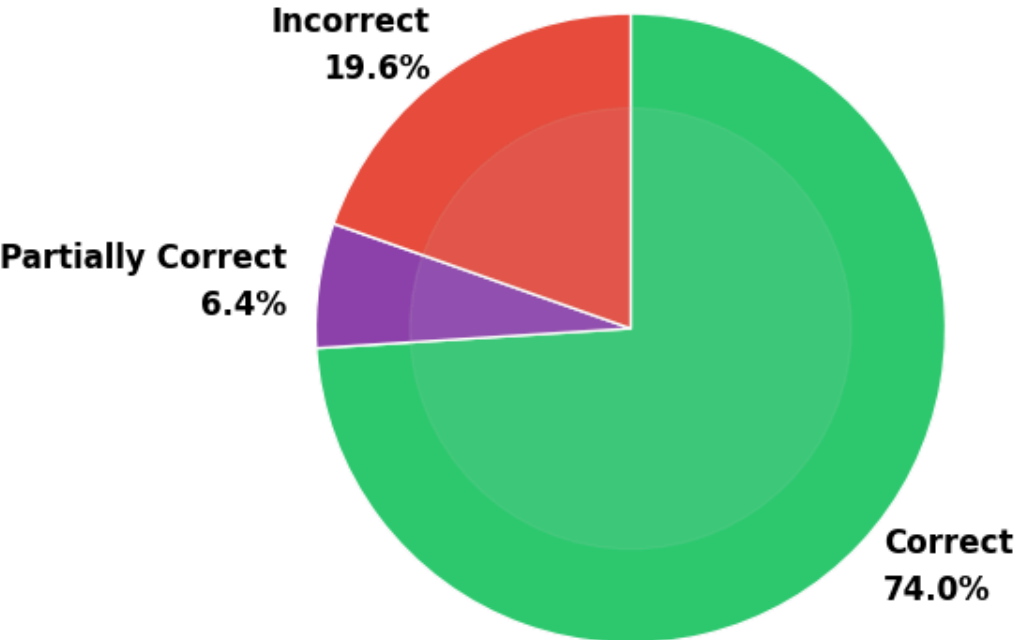


Error Analysis: Task 2

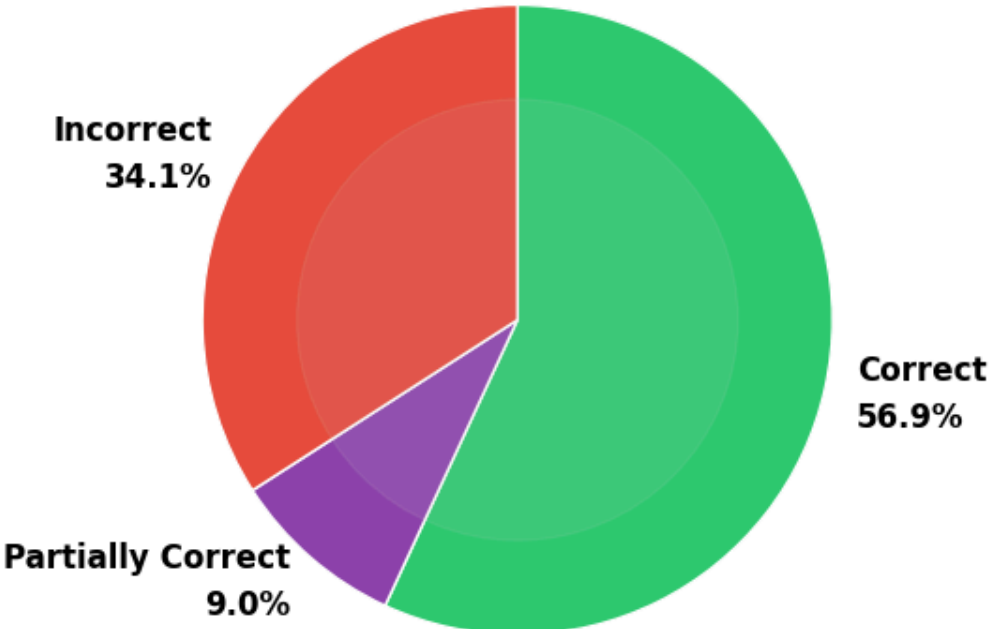
ChatGPT



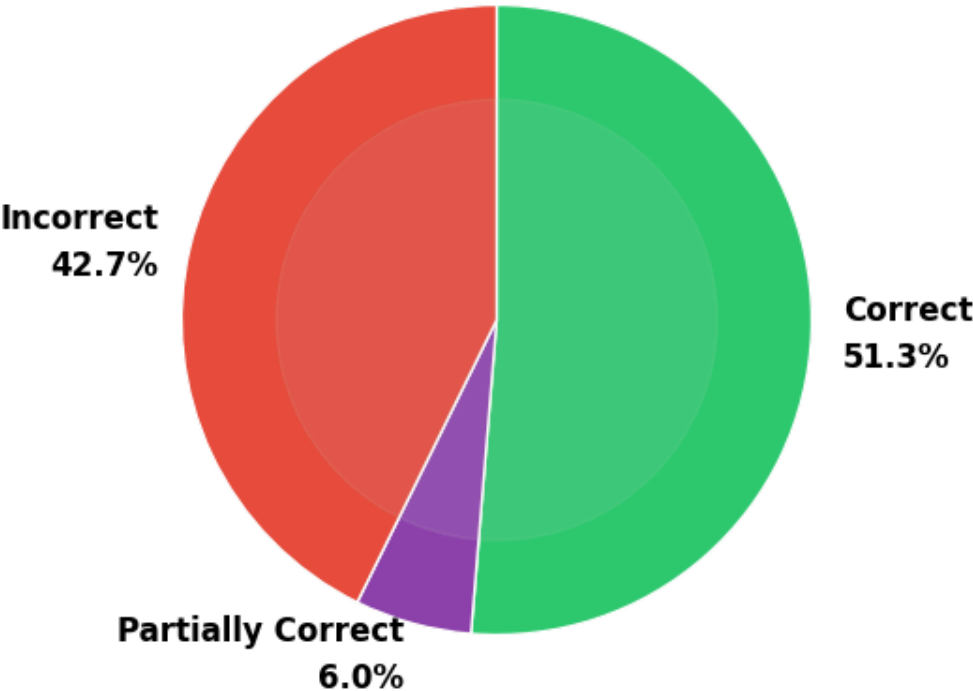
Gemini



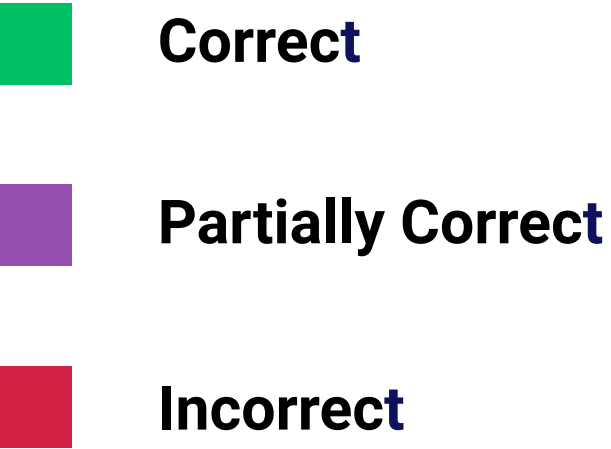
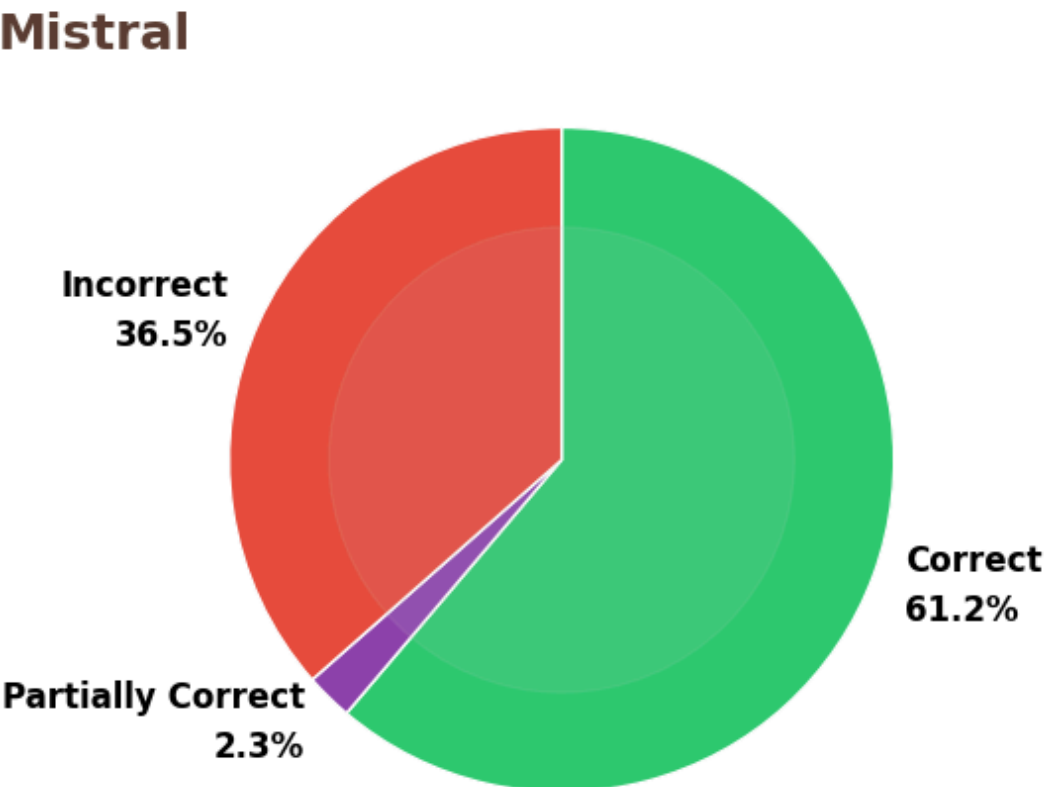
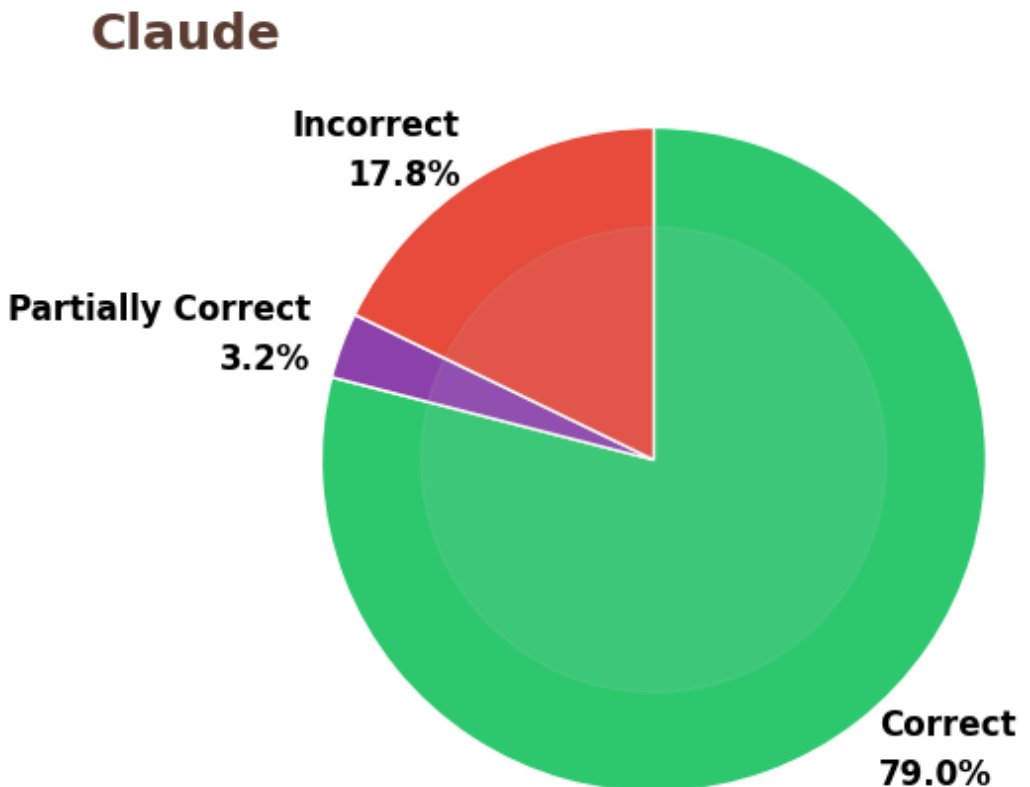
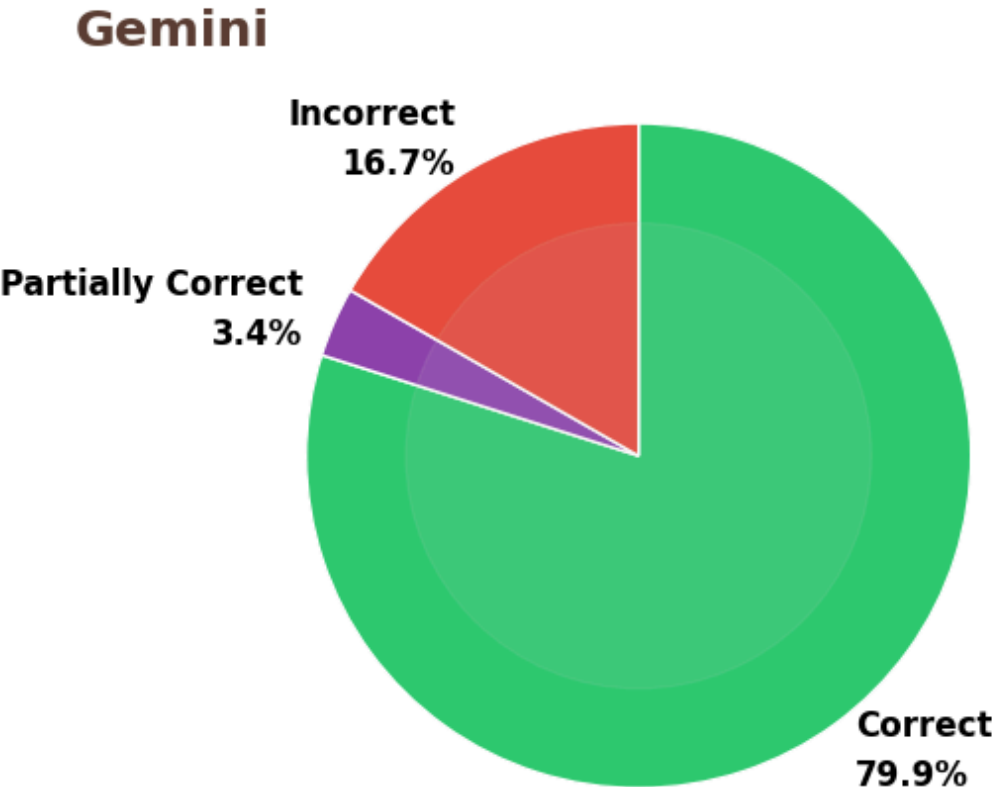
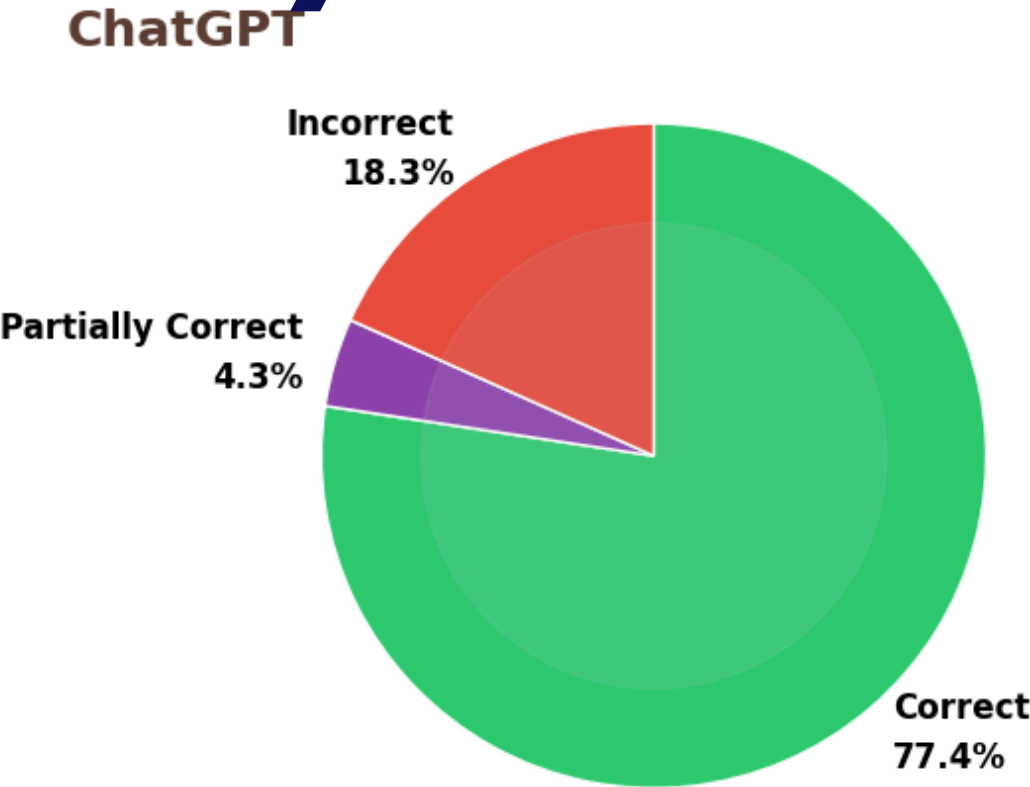
Claude



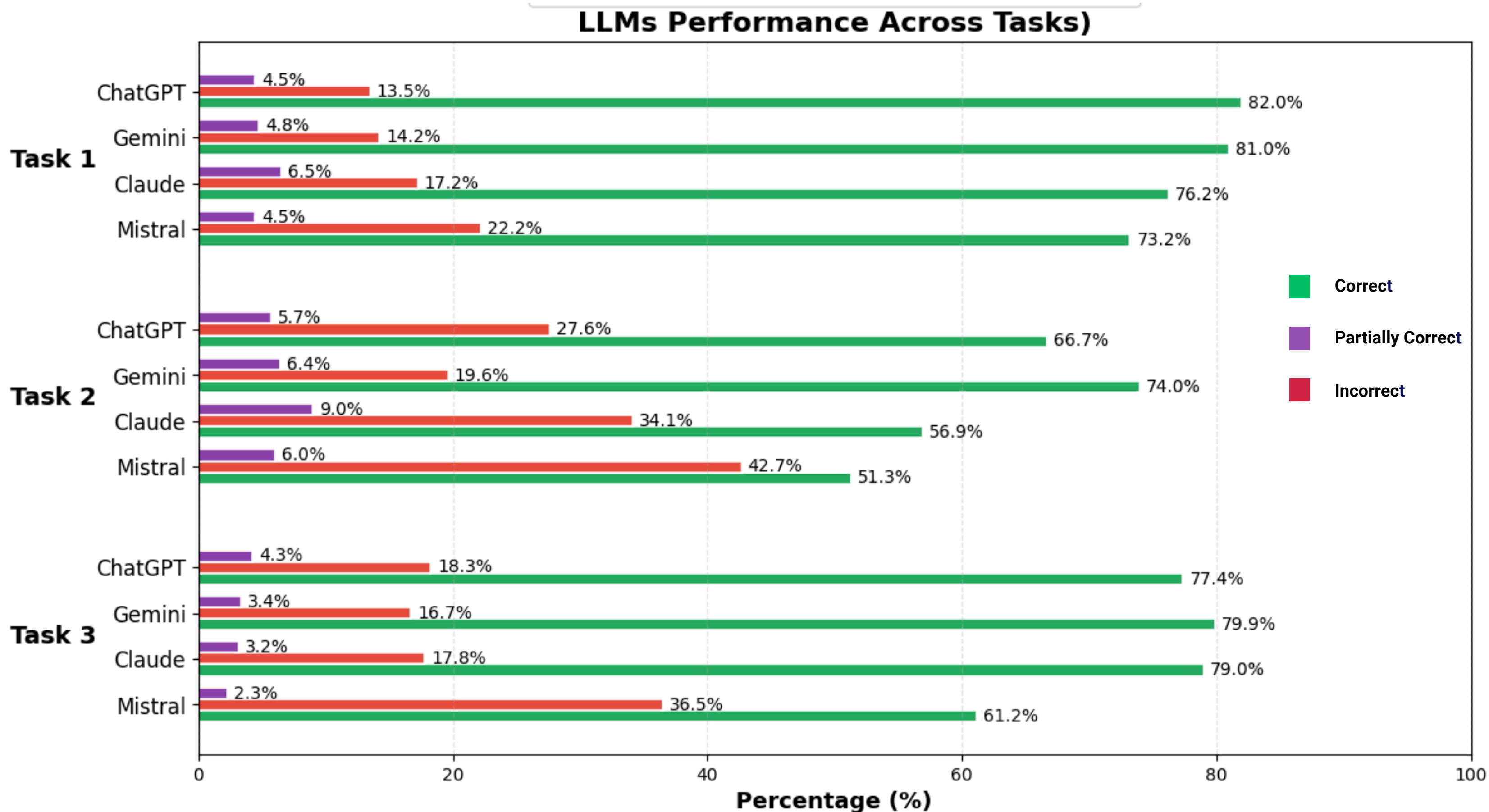
Mistral



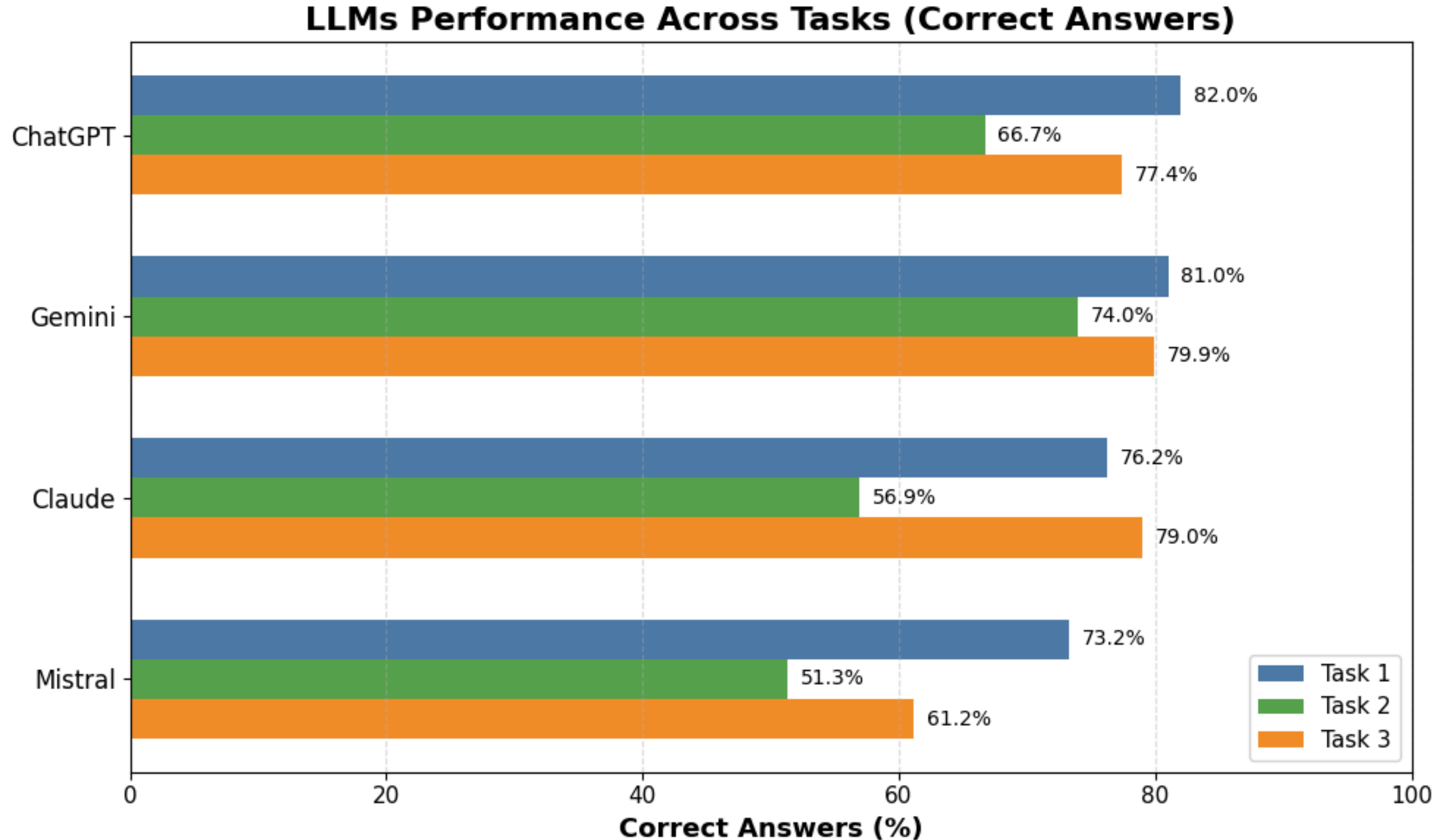
Error Analysis: Task 3



Summary of Results Across Tasks



Summary of Results Across Tasks (Correct Answer)



Conclusion

Task 1: Visual Encoding Interpretation

- Top Performers: ChatGPT (82%) and Gemini (81%)

Task 2: Chart-Based Question Answering

- Top Performer: Gemini (74%)

Task 3: Detection of Misleading Charts

- Top Performers: Gemini (79.9%) and Claude (79%)

Overall across all tasks:

- Gemini = most consistent top performer
- ChatGPT = strong and stable across all categories
- Claude & Mistral = more variable, especially in reasoning-heavy tasks

LLMs perform worst on reasoning-heavy tasks (Task 2), compared to chart interpretation (Task 3) and basic perception (Task 1). They performde best in Taks 1.

The results show that LLMs perform poorest on reasoning-intensive tasks, indicating persistent limitations in multi-step numerical and logical reasoning compared to visula encoding and misleading charts.

Future Work

1. Expanded Evaluation

- Include more chart types, diverse task categories, and datasets with varying levels of complexity for better generalization.

2. Improved Metrics

- Develop better scoring methods that assess partial correctness, reasoning quality, and explanation clarity.

3. Robust Testing

- Increase test cases and add complex or adversarial scenarios to better evaluate model limits.

4. Prompt Engineering

- Design structured, step-by-step prompts to improve accuracy, consistency, and reduce hallucinations in visualization tasks.

Thank You!

References

1. <https://www.linkedin.com/pulse/types-charts-data-visualization-comprehensive-guide-krithik-s-b0fhc>
2. Donut chart: <https://www.limesurvey.org/blog/knowledge/how-and-when-to-use-these-most-popular-chart-types-for-your-survey>
3. https://infovis-wiki.net/wiki/Visual_Variables